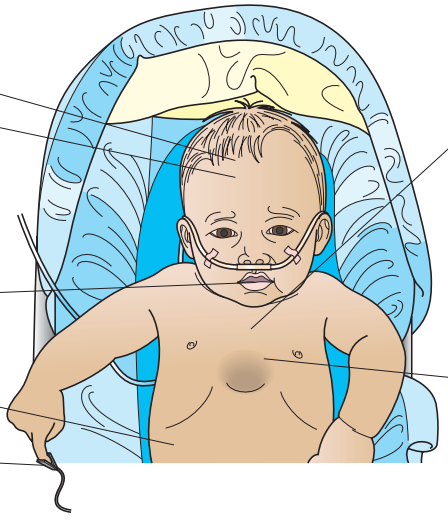
Summary

The infant with tachypnoea or wheeze

Clinical features to assess



Clinical conditions to consider

Bronchiolitis
- Age 1–9 months
- Poor feeding, apnoea, dry cough
- Laboured breathing – chest recession, hyperinflation of the chest, fine end-inspiratory crackles, wheeze, liver displaced downwards
- Apnoea, cyanosis, respiratory failure
- Increased severity with bronchopulmonary dysplasia in preterm or congenital heart disease

Pneumonia
- Fever, poor feeding, cough, lethargy, cyanosis
- Tachypnoea, nasal flaring, chest recession, wheeze and end-inspiratory coarse crackles over the affected area
- O_2 saturation may be decreased
- Chest X-ray – consolidation, parapneumonic effusion or empyema

Transient early wheezing – with viral infections, risk increased in preterm and maternal smoking

Non-atopic wheezing – following viral lower respiratory infection

Atopic asthma – recurrent wheezing, eczema, positive family history of allergy/atopy

Cardiac failure – respiratory distress, heart murmur, hepatomegaly

Inhaled foreign body – choking on peanut or toy, etc.
Aspiration of feeds – especially with neuromuscular disorder
Other causes – see Box 16.2

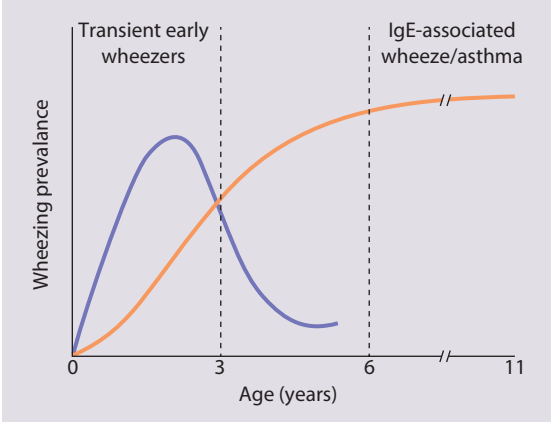


Figure 16.9 Prevalence of wheeze in children caused by the two major phenotypes by age.

condition its episodic nature, being triggered by viruses that cause the common cold. Studies have found that transient early wheezers often have decreased lung function from birth, from small airway diameter. Risk factors include maternal smoking during and/or after pregnancy and prematurity. A family history of asthma or allergy is not a risk factor. Transient early wheezing is more common in males and usually resolves by 5 years of age, presumably from the increase in airway size.

Persistent and recurrent wheezing

Some children, both preschool and school-aged, have frequent wheeze triggered by many stimuli. The presence of IgE to common inhalant allergens, such as house dust mite, pollens or pets, is associated with persistence of wheezing beyond the preschool years. Recurrent wheezing associated with evidence of allergy to one or more inhaled allergens (e.g. by skin-prick test or IgE blood test) is termed 'atopic asthma'. Atopic wheezers have persistent symptoms and decreased lung function. Atopic asthma is strongly associated with other atopic diseases such as eczema, rhinoconjunctivitis and food allergy, and is more common in those with a family history of such diseases.

Box 16.2 Causes of childhood wheeze

- Transient early wheezing
- Atopic asthma (IgE-mediated)
- Non-atopic asthma
- Recurrent aspiration of feeds
- Inhaled foreign body
- Cystic fibrosis
- Recurrent anaphylaxis in a child with food allergies
- Congenital abnormality of lung, airway or heart
- Idiopathic.

A small number of persistent or recurrent wheezing children will have other causes, such as the non-atopic asthmatics. Other causes of recurrent wheeze are listed in Box 16.2.

Pathophysiology of asthma

An outline of the pathophysiology of asthma is shown in Figure 16.10.

Up to 40% of all children are atopic (see Ch. 15). Siblings and parents have an increased risk of allergic diseases. The presence of one allergic condition increases the risk of another, e.g. half of children with allergic asthma have eczema at some time during their lives. The majority of asthma exacerbations are triggered by rhinovirus infection, and there is evidence that those with asthma have specific immune defects which increase their vulnerability to these viruses.

Clinical features

Asthma should be suspected in any child with wheezing on more than one occasion. Wheeze is a polyphonic (multiple pitch) noise coming from the airways believed to represent many airways of different dimensions vibrating from abnormal narrowing. Although it may be clear to most clinicians what 'wheezing' is, patients and parents do not always mean the same thing. It is best to describe the sound to a parent (e.g. 'a whistling in the chest when your child breaths out') and ask if that fits with their child's symptoms. Ideally, the presence of wheeze is confirmed on auscultation by a health professional to distinguish it from transmitted upper respiratory noises. Other key features associated with a high probability of a child having asthma include:

- Symptoms worse at night and in the early morning
- Symptoms that have triggers (e.g. exercise, pets, dust, cold air, emotions, laughter)
- Interval symptoms, i.e. symptoms between acute exacerbations

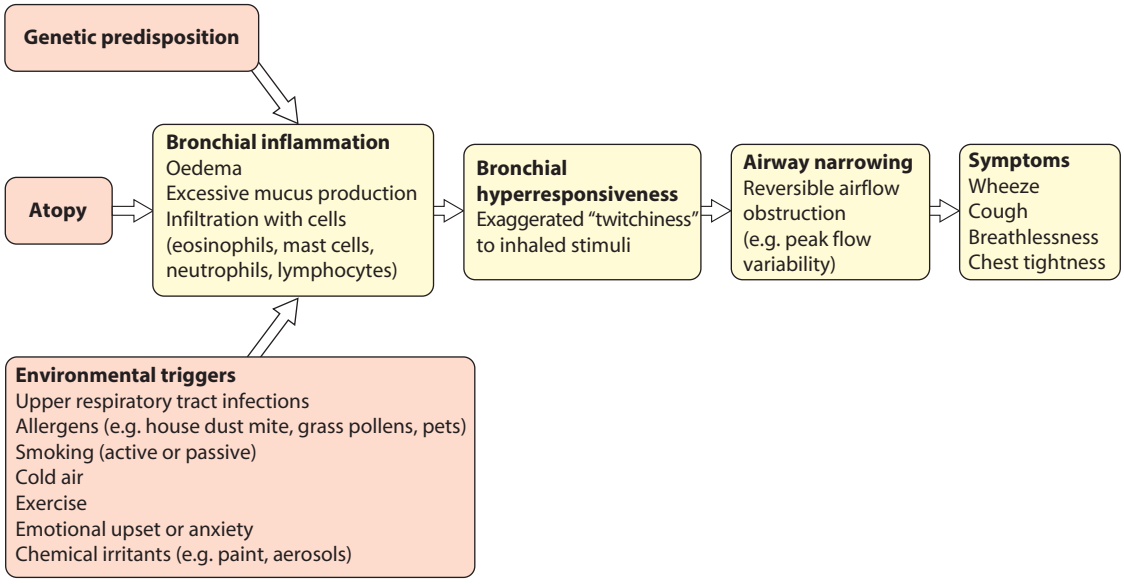


Figure 16.10 Pathophysiology of asthma.

- Personal or family history of an atopic disease
- Positive response to asthma therapy.

Once suspected, the pattern or phenotype should be further explored by asking:

- How frequent are the symptoms?
- What triggers the symptoms? Specifically, are sport and general activities affected by the asthma?
- How often is sleep disturbed by asthma?
- How severe are the interval symptoms between exacerbations?
- How much school has been missed due to asthma?

Examination of the chest is usually normal between attacks. In long-standing asthma there may be hyperinflation of the chest, generalised polyphonic expiratory wheeze and a prolonged expiratory phase. Onset of the disease in early childhood may result in Harrison sulci (Fig. 16.11). Evidence of eczema should be sought, as should examination of the nasal mucosa for allergic rhinitis. Growth should be plotted but is normal unless the asthma is extremely severe. The presence of a wet cough or sputum production, finger clubbing, or poor growth suggests a condition characterised by chronic infection such as cystic fibrosis or bronchiectasis.

In practice, the diagnosis is usually made on a history of recurrent wheeze, with exacerbations usually precipitated by viral respiratory infections.



Figure 16.11 The depressions at the base of the thorax associated with the muscular insertion of the diaphragm are called Harrison sulci, and are associated with chronic obstructive airways disease such as asthma during childhood.

Investigations

Asthma can usually be diagnosed from the history and examination and no investigations are needed. Sometimes, specific investigations are required to confirm the diagnosis, or explore the severity and phenotype in more detail. Skin-prick testing for common allergens is often considered both as an aid to the diagnosis of atopy and to identify allergens which may be acting as triggers. A chest X-ray is usually normal but may help to rule out other conditions. If there is uncertainty, recording peak expiratory flow rate (PEFR) may be useful. Most children over 5 years of age can use a peak flow meter. Uncontrolled asthma leads to increased variability in peak flow, with both diurnal variability (morning PEFR usually lower than evening PEFR) and day-to-day variability (change in PEFR over the course of a week). Often, response to treatment is the most helpful investigation. This can be assessed if

Summary

Assessment of the child with chronic asthma

Clinical features to check

Growth and nutrition

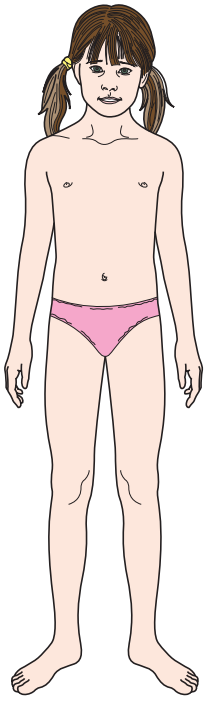
Peak flow/spirometry

Chest for:
Hyperinflation
Harrison's sulcus
Wheeze

Are there other allergic disorders?
- Allergic rhinitis
- Eczema, etc.

If there is:
- Sputum
- Finger clubbing
- Growth failure

– If present, other causes should be sought



Monitor:
- Peak flow diary
- Severity and frequency of symptoms
- Exercise tolerance
- Interference with life, time off school
- Is sleep disturbed?
- Use of preventer and reliever medication – are they appropriate?
- Inhaler technique

Consider triggers:
- Allergic rhinitis needing treatment?
- Allergens - animal dander, etc.
- Stress

Table 16.2 Drugs in asthma

Type of drug	Drug
Bronchodilators	
β_2-agonists (relievers)	Salbutamol
	Terbutaline
Anticholinergic bronchodilator	Ipratropium bromide
Preventative/prophylactic treatment	
Inhaled steroids	Budesonide
	Beclometasone
	Fluticasone
	Mometasone
Long-acting β_2-bronchodilators	Salmeterol
	Formoterol
Methylxanthines	Theophylline
Leukotriene inhibitors	Montelukast
Oral steroids	Prednisolone
Anti-IgE injections	Omalizumab

All are given by inhalation, except prednisolone, leukotriene modulators, theophylline preparations, which are by mouth, and omalizumab, which is by injection.

necessary by measuring the PEFR before and after inhaling a bronchodilator; it should increase by more than 10–15%.

Management

The aim of management is to allow the child to lead as normal a life as possible by controlling symptoms and preventing exacerbations, optimising pulmonary function, while minimising treatment and side-effects. It is important to set the aims with the child as they are more likely to comply with their therapy if they are involved in their management. An evidence-based and regularly updated British Guideline on Asthma Management gives guidance on asthma treatment in children and adults.

Medications used to treat children with asthma are shown in Table 16.2.

Bronchodilator therapy

Inhaled β_2-agonists are the most commonly used and most effective bronchodilators. *Short-acting β_2-agonists* (often called *relievers*) such as salbutamol or terbutaline have a rapid onset of action, are effective for 2–4 h and have few side-effects. They are used as required for increased symptoms, and in high doses for acute asthma attacks.

In contrast, *long-acting β_2-agonists* (*LABAs*) such as salmeterol or formoterol are effective for 12 h. They are not used in acute asthma, and should not be used without an inhaled corticosteroid. LABAs are useful in exercise-induced asthma.

Ipratropium bromide, an anticholinergic bronchodilator, is sometimes given to young infants when other bronchodilators are found to be ineffective, or in the treatment of severe acute asthma.

Inhaled corticosteroids

Prophylactic drugs are effective only if taken regularly. *Inhaled corticosteroids* (often called *preventers*) are the most effective inhaled prophylactic therapy. They decrease airway inflammation, resulting in decreased symptoms, asthma exacerbations and bronchial hyperactivity. They are increasingly used in conjunction with an inhaled LABA. They have no clinically significant side-effects when given in conventional licensed doses. They can produce systemic side-effects, including impaired growth, adrenal suppression and altered bone metabolism, when high doses are used.

Add-on therapy

The first choice of add-on therapy in a child over 5 years is a LABA, whereas in children under 5 years, an oral *leukotriene receptor antagonist* such as montelukast is recommended. The latter can also be used in older children when symptoms are not controlled by the addition of a LABA. *Slow-release oral theophylline* is an alternative; however, it has a high incidence of side-effects (vomiting, insomnia, headaches, poor concentration), so it is not commonly used in children.

Other therapies

Oral *prednisolone*, usually given on alternate days to minimise the adverse effect on height, is required only in severe persistent asthma where other treatment has failed. All children on this therapy must be managed by a specialist in childhood asthma. *Anti-IgE therapy* (*omalizumab*) is an injectable monoclonal antibody that acts against IgE, the natural antibody that mediates allergy. It is used for the treatment of severe atopic asthma, and should only be administered by a specialist in childhood asthma.

Most antibiotics are of no value in the absence of a bacterial infection, although recent data suggest that macrolide antibiotics (e.g. erythromycin) may have a specific role in asthma management, but neither cough medicines nor decongestants are helpful. Antihistamines, e.g. loratadine and nasal steroids, are useful in the treatment of allergic rhinitis.

The British Guideline on Asthma Management uses a stepwise approach, starting treatment with the step most appropriate to the severity of the asthma. Treatment increases from step 1 (mild intermittent asthma) to step 5 (chronic severe asthma requiring continuous or frequent use of oral steroids), stepping down when control is good (Fig. 16.12).

Allergen avoidance and other non-pharmacological measures

Although asthma in many children is precipitated or worsened by specific allergens, complete avoidance of the allergen is difficult to achieve. The value of identifying such triggers by history or allergy testing is controversial. There is currently no conclusive evidence that allergen avoidance measures (such as removal of furry

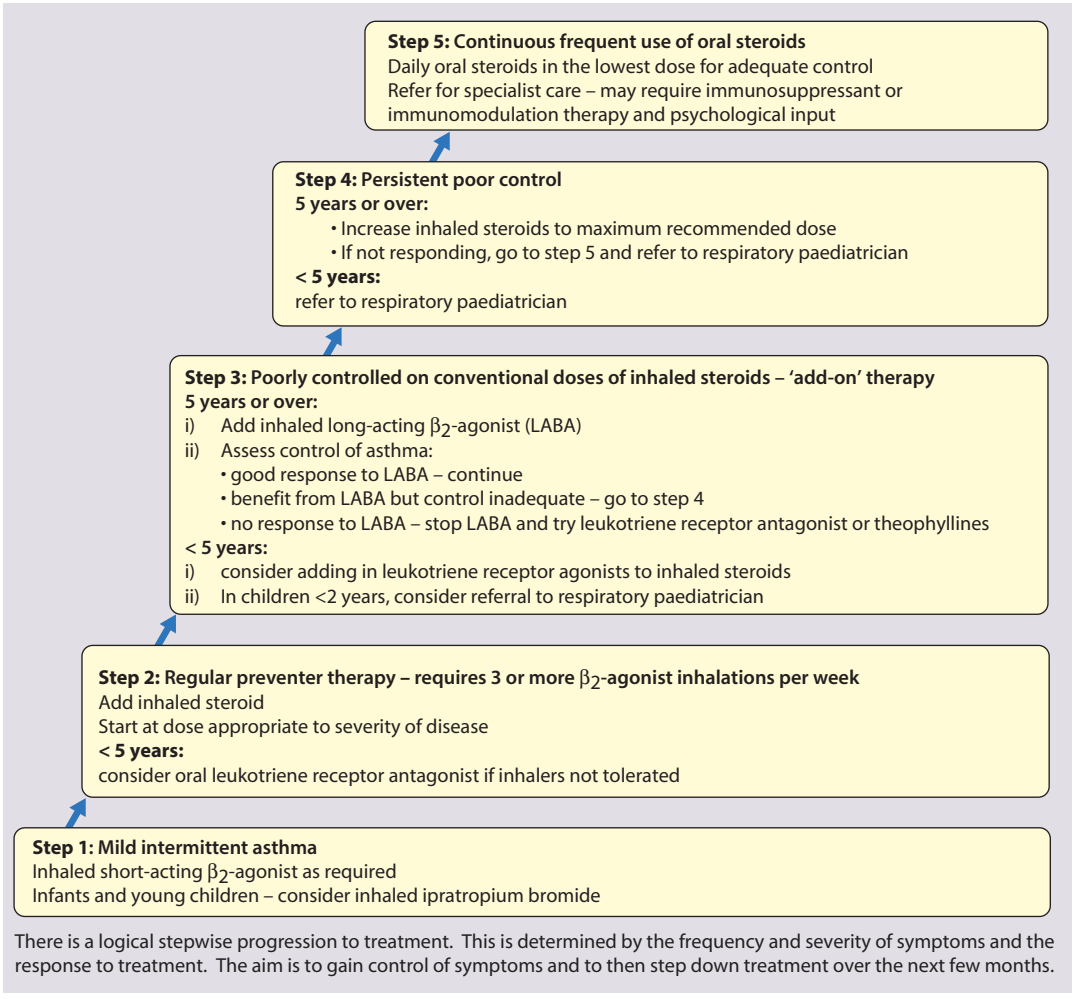


Figure 16.12 A stepwise approach to the treatment of asthma. (From: British Thoracic Society/Scottish Intercollegiate Guideline Network. 2009. The British Guideline on Asthma Management. Reproduced with permission. Available at: http://www.sign.ac.uk/guidelines/ (Accessed January 2011).)

animals or using dust mite impermeable mattress covers) are beneficial, although they may be considered in selected cases. Allergen immunotherapy is effective for treating atopic asthma, but its use is limited by the risk of systemic allergic reactions associated with the treatment (see Ch. 15).

Parents should be advised about the harmful effects of cigarette smoking in the house. Although exercise improves general fitness, there is no evidence that physical training improves asthma itself. Psychological intervention may be useful in chronic severe asthma.

Exercise-induced asthma

Some children's asthma is brought on only by vigorous exercise. With appropriate treatment, asthma should not restrict exercise, and there are many elite athletes with asthma. For most, a short-acting β_2-agonist bronchodilator taken immediately before exercise is sufficient, but if there are more marked symptoms a LABA taken in conjunction with an inhaled steroid will give greater protection. A LABA should not be prescribed without an inhaled steroid.

Acute asthma

Assessment

With each acute attack, the duration of symptoms, the treatment already given and the course of previous attacks should be noted. Clinical features are:

- Wheeze and tachypnoea (respiratory rate >50 breaths/min in children 2–5 years, >30 breaths/min in children ≥5 years) – but poor guide to severity
- Increasing tachycardia (>130 beats/min in children aged 2–5 years, >120 beats/min in children ≥5 years) – better guide to severity
- The use of accessory muscles and chest recession – also better guide to severity
- The presence of marked pulsus paradoxus (the difference between systolic pressure on

Choosing the correct inhaler

Inhaled drugs may be administered via a variety of devices, chosen according to the child's age and preference:

- *Pressurised metered dose inhaler (pMDI) and spacer* (Fig. 16.13).
  - Appropriate for all age groups: 0–2 years, spacer and facemask; >2 years, spacer alone
  - A spacer is recommended for all children as it increases drug deposition to the lungs
  - Useful for acute asthma attacks when poor inspiratory effort may impair the use of inhalers direct to the mouth
- *Breath-actuated metered dose inhalers* (e.g. *Autohaler*, *Easibreath*): 6+ years. Less coordination needed than a pMDI without spacer. Useful for delivering β-agonists when 'out and about' in older children
- *Dry powder inhaler*: 4+ years (Fig. 16.14). Needs a good inspiratory flow, therefore less good in severe asthma and during an asthma attack. Also easy to use when 'out and about' in older children
- *Nebuliser*: any age (Fig. 16.15). Only used in acute asthma where oxygen is needed in addition to inhaled drugs; occasionally used at home as part of an acute management plan in those with rapid-onset severe asthma (brittle asthma).

Many children fail to gain the benefit of their treatment because they cannot use the inhaler correctly. This must be demonstrated and the child's ability to use it checked. In young children, parents need to be skilled in assisting their child to use the inhaler correctly. Assessing and reassessing inhaler technique is vital to good management and should be a routine part of any review.



Figure 16.13 Pressurised metered dose inhaler (pMDI) and spacer. Suitable for all ages, with facemask if <2 years old.



Figure 16.14 Dry powder inhaler, >4 years.



Figure 16.15 Nebuliser: all ages. Only used in acute asthma where oxygen is needed in addition to inhaled drugs.

inspiration and expiration) indicates moderate to severe asthma attack in children but is difficult to measure accurately and is therefore unreliable
- If breathlessness interferes with talking, the attack is severe
- Cyanosis, fatigue and drowsiness are late signs, indicating life-threatening asthma; this may be accompanied by a silent chest on auscultation as little air is being exchanged. This is an emergency as the child may be about to arrest.

However, the severity of an acute asthma attack may be underestimated by clinical examination alone. Therefore:

- Arterial oxygen saturation should be measured with a pulse oximeter in all children presenting to hospital with acute asthma. Oxygen saturation <92% in air implies severe or life-threatening asthma
- Measurement of the peak expiratory flow rate should be routine in school-age children.

The features of a severe and life-threatening acute attack are shown in Figure 16.16.

Criteria for admission to hospital

Children require hospital admission if, after high-dose inhaled bronchodilator therapy, they:

- have not responded adequately clinically; there is persisting breathlessness, tachypnoea
- are exhausted
- still have a marked reduction in their predicted (or usual) peak flow rate
- have a reduced oxygen saturation (<92% in air).

A chest X-ray is indicated only if there are unusual features (e.g. asymmetry of chest signs suggesting pneumothorax, lobar collapse) or signs of severe infection. In children, blood gases are only indicated in life-threatening or refractory cases.

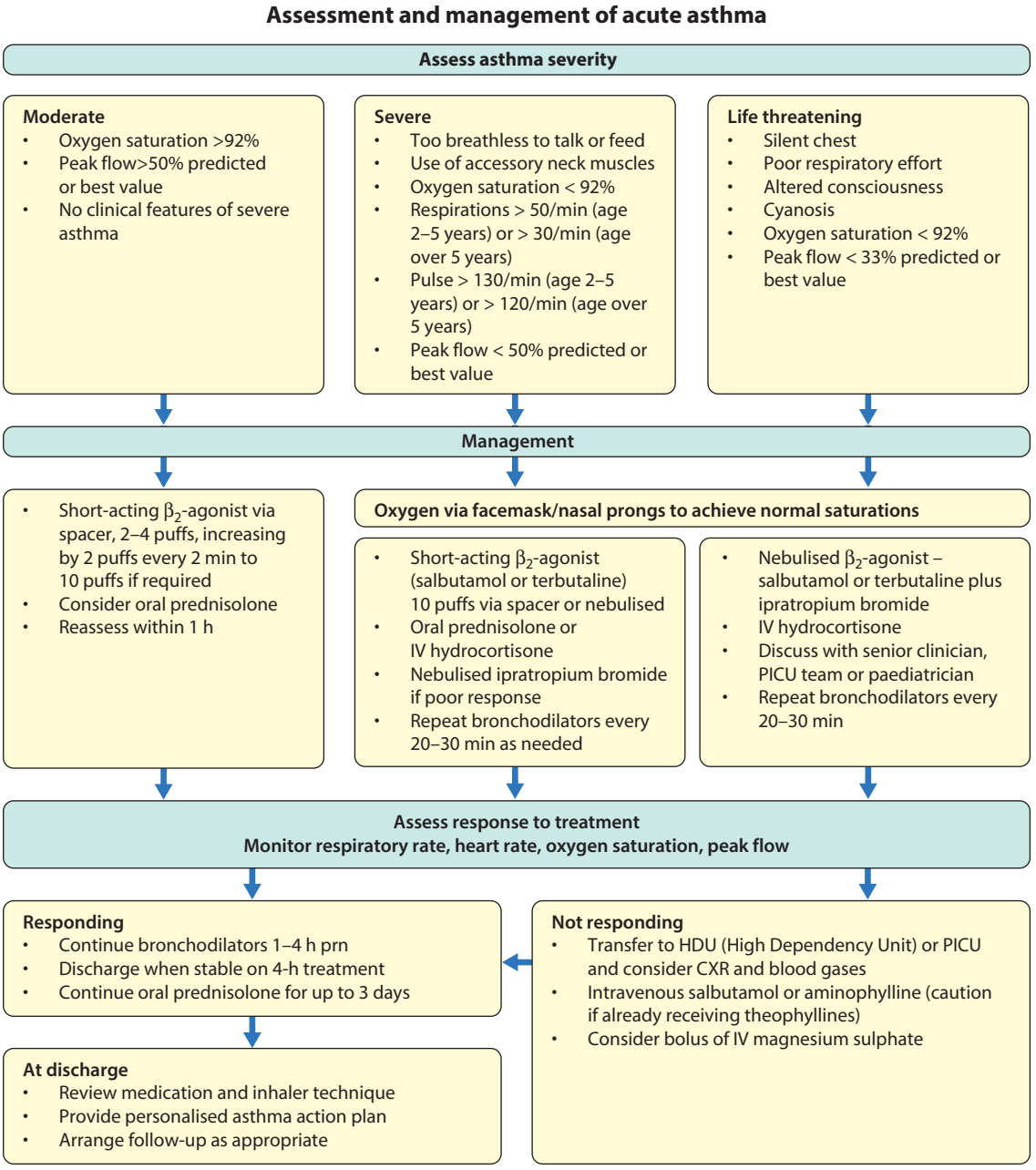


Figure 16.16 Assessment and management of acute asthma. (Adapted from the British Guideline on Asthma Management, 2003.)

Management

Acute breathlessness is frightening for both the child and the parents. Calm and skilful management is the key to their reassurance. High-dose inhaled bronchodilators, steroids and oxygen form the foundation of therapy of severe acute asthma.

Management is summarised in Figure 16.16. As soon as the diagnosis has been made, the child should be given a β_2-bronchodilator. For severe exacerbations, high-dose therapy should be given and repeated every 20–30 min. For moderate to severe asthma, 10 puffs of β_2-bronchodilator should be given via a pressurised metered dose inhaler (pMDI) and large volume spacer. This is not only good treatment but also educates the child and parent in using the preferred devices they will have at home. For severe to life-threatening asthma, a β_2-bronchodilator may need to be given via nebuliser driven by high-flow oxygen. The addition of nebulised ipratropium to the initial therapy in severe asthma is beneficial. Oxygen is given when there is any evidence of arterial oxygen desaturation, such as saturations of <92%. A short course (2–5 days) of oral prednisolone expedites the recovery from moderate or severe acute asthma.

Intravenous therapy has a role in the minority of children who fail to respond adequately to inhaled bronchodilator, either aminophylline or intravenous

Summary

Assessment of the child with acute asthma

Determine the severity of the attack
(see Fig 16.16)
- Mild
- Moderate
- Severe
- Life-threatening

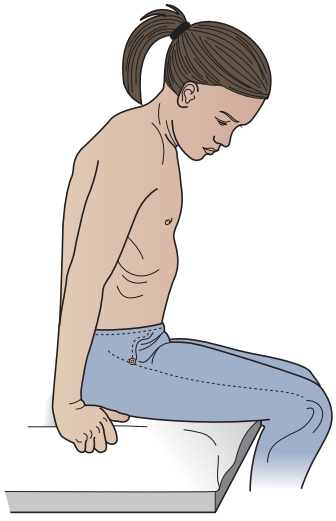
Too breathless to talk or eat?

Increased work of breathing
- Tachypnoea – severe if >30 breaths/min

Chest recession:
- Moderate – some intercostal recession
- Severe – use of accessory neck muscles
- Life-threatening – poor respiratory effort

Auscultation:
- Wheeze
- Silent chest – poor air entry in life-threatening

Pulse:
- Severe – >120 beats/min



Level of consciousness – altered in life-threatening
Exhaustion

Tongue:
- Cyanosis in life-threatening

Peak flow (% predicted):
- Moderate >50%
- Severe <50%
- Life-threatening <33%

O_2 saturation:
- Moderate >92%
- Severe or life-threatening <92%

Is there a trigger for the attack?:
- URTI or other viral illness
- Pneumonia
- Allergen, e.g. animal dander
- Exercise
- Cold air

salbutamol. For intravenous aminophylline, a loading dose is given over 20 min, followed by continuous infusion. Seizures, severe vomiting and fatal cardiac arrhythmias may follow a rapid infusion. If the child is already on oral theophylline, the loading dose should be omitted. With both aminophylline and salbutamol, the ECG should be monitored and blood electrolytes checked. There is increasing evidence that intravenous magnesium sulphate is helpful in life-threatening asthma. Antibiotics are only given if there are clinical features of bacterial infection. Occasionally, these measures are insufficient and artificial ventilation is required.

Patient education

Prior to discharge from hospital after an acute admission, the following points should be reviewed with the child and family:

- When drugs should be used (regularly or 'as required')
- How to use the drug (inhaler technique)
- What each drug does (relief vs prevention)
- How often and how much can be used (frequency and dosage)
- What to do if asthma worsens (a personal asthma management plan is helpful: see the Appendix for an example).

The child and parents need to know that increasing cough, wheeze and breathlessness, and difficulty in walking, talking and sleeping, or decreasing relief from bronchodilators all indicate poorly-controlled asthma. Some asthmatics find it difficult to identify gradual deterioration – measurement of peak flow rate at home allows earlier recognition. Patients with troublesome asthma are usually given a supply of oral steroids to keep at home, with instructions in the asthma action plan on when to start them.

Recurrent or persistent cough

Cough is the most common symptom of respiratory disease and indicates stimulation of nerve receptors in the pharynx, larynx, trachea or large bronchi. For most children, episodes of cough are due to upper respiratory tract infections caused by the common cold viruses and do not indicate the presence of a long-term or serious underlying respiratory disease. Cough appears persistent because of a series of respiratory tract infections, although some infections, such as pertussis, RSV and *Mycoplasma* infection, can cause a cough that persists for weeks or months. The challenge for the physician is to identify children with other, less common, clinically significant causes of recurrent or persistent cough (Box 16.3).

Asthma is the next most common cause of recurrent cough in childhood. Although there is usually associated wheeze and breathlessness, sometimes the wheezing is not recognised or not described

Box 16.3 Causes of recurrent or persistent cough

- Recurrent respiratory infections
- Post-specific respiratory infections (e.g. pertussis, RSV, *Mycoplasma*)
- Asthma
- Suppurative lung diseases (e.g. cystic fibrosis, ciliary dyskinesia or immune deficiency)
- Recurrent aspiration (± gastro-oesophageal reflux)
- Persistent endobronchial infection
- Inhaled foreign body
- Cigarette smoking (active or passive)
- Tuberculosis
- Habit cough
- Airway anomalies (e.g. tracheo-bronchomalacia, tracheo-oesophageal fistula).

accurately. Identifying wheeze on auscultation during an acute episode is helpful to make the diagnosis. However, many children with persistent cough without wheeze are treated incorrectly as asthmatics. If the clinical features are not suggestive of asthma or if initial treatment is not beneficial, other diagnoses should be considered or the child referred to a paediatrician with a specialist interest in respiratory disorders.

Persistent cough after an acute infection may indicate cystic fibrosis or unresolved lobar collapse, which will be seen on a chest X-ray. Most children will not expectorate sputum but will swallow their sputum. It is therefore crucial to listen to the quality of the cough. If 'wet' (i.e. sounding like there is excess sputum in the airways) or if the cough is productive, further investigation is required (see below). In any child with a severe, persistent cough, TB should be excluded with a chest X-ray and tuberculin skin (Mantoux) test.

Aspiration of feeds may cause cough and wheeze. This may be caused by gastro-oesophageal reflux or as a result of swallowing disorders, e.g. in children with cerebral palsy. Inhaled foreign body needs to be considered even when there is no clear history.

The significance of parental smoking on children is generally underestimated. If both parents smoke, young children are twice as likely to have recurrent cough and wheeze than in non-smoking households. In the older child, active smoking is common: 10% of 11–15-year-olds and 30% of 16–19-year-olds smoke regularly.

Some older children and adolescents develop a barking, unproductive, habit cough following an infection or an asthma attack. The cough characteristically disappears during sleep and is dry in nature. Reassurance and explanation after a thorough examination are usually effective.

Chronic lung infection

Any child with a persistent cough that sounds 'wet' (i.e. sounds like there is excess sputum in the chest) or is productive should be investigated. The child may have bronchiectasis, permanent dilatation of the bronchi. Bronchiectasis may be generalised or restricted to a single lobe. Generalised bronchiectasis may be due to cystic fibrosis, primary ciliary dyskinesia, immunodeficiency or chronic aspiration. Cystic fibrosis is considered separately below. Focal bronchiectasis is due to previous severe pneumonia, congenital lung abnormality or obstruction by a foreign body (see Case History 16.2).

In primary ciliary dyskinesia there is congenital abnormality in the structure or function of cilia. This leads to impaired mucociliary clearance. Affected children have recurrent infection of the upper and lower respiratory tract, which if untreated may lead to severe bronchiectasis. They characteristically have a recurrent productive cough, purulent nasal discharge and chronic ear infections; 50% also have dextrocardia and situs inversus (Kartagener syndrome). The diagnosis is made in a specialist laboratory by examination of the structure and function of the cilia of nasal epithelial cells brushed from the nose. The cornerstones of management are daily physiotherapy to clear secretions, proactive treatment of infections with antibiotics and appropriate ENT follow-up.

Children with immunodeficiency may develop severe, unusual or recurrent chest infections. The immune deficiency may be secondary to an illness, e.g. malignant disease or its treatment with chemotherapy. Less commonly it is due to HIV infection or a primary immune deficiency.

Many children with neurodisability will have chronic aspiration, either due to oropharyngeal incoordination or due to gastro-oesophageal reflux.

Tuberculosis remains an important cause of chronic lung infection and all children with a persistent productive cough should have a chest X-ray and tuberculin skin test. Marked hilar or paratracheal lymphadenopathy is highly suggestive of tuberculosis.

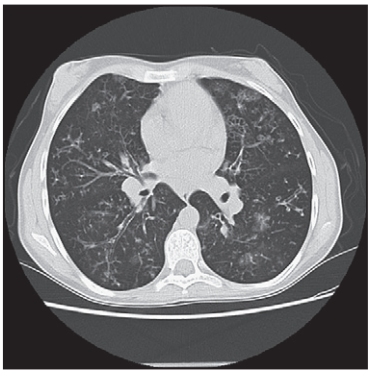
Persistent inflammation of the lower airways driven by chronic infection of the lower respiratory tract (persistent endobronchial infection) is increasingly recognised as a cause of chronic wet cough in children. It may be a precursor to bronchiectasis if investigations and treatment are not instituted. Referral to a specialist in paediatric respiratory disorders is indicated. Persistent endobronchial infection is often improved with early access to oral antibiotics or on occasions long-term prophylactic antibiotics.

A plain chest X-ray may show gross bronchiectasis, but will often not identify it. Bronchiectasis is best identified on a CT scan of the chest (Fig. 16.17a,b). To investigate focal disease bronchoscopy is usually indicated to exclude a structural cause.

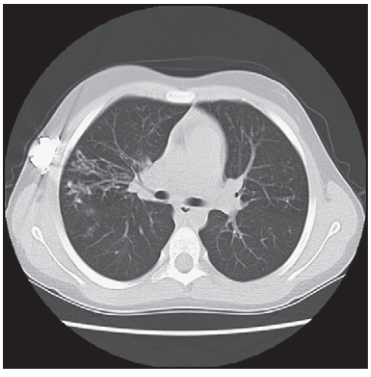
Cystic fibrosis

Epidemiology, genetics and basic defect

Cystic fibrosis (CF) is the commonest life-limiting autosomal recessive condition in Caucasians with an incidence of 1 in 2500 live births and carrier rate of 1 in 25.



(a)



(b)

Figure 16.17 Bronchiectasis on CT scan of the chest. **(a)** Generalised and **(b)** focal, in the right upper lobe.

Case History

16.2 Foreign body inhalation

A previously well 3-year-old boy presented with a 5-day history of severe cough and wheeze. His symptoms developed after choking on some peanuts. A chest X-ray revealed a hyperlucent right lung (Fig. 16.18). Bronchoscopy was performed and revealed a peanut wedged in the right main bronchus.

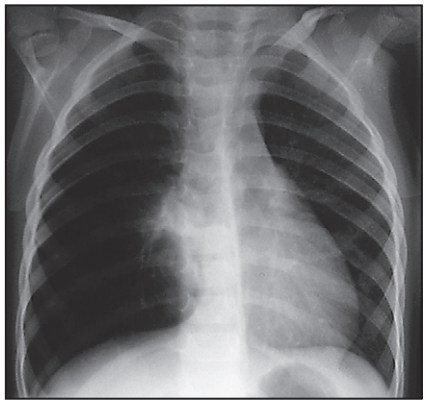


Figure 16.18 Hyperlucency of the right lung and mediastinal shift to the left. (Courtesy of Dr Abbas Khakoo.)

It is well recognised but less common in other ethnic groups. Average life expectancy has increased from a few years to the mid-30s, with a projected life expectancy for current newborns into the 40s.

The fundamental problem in CF is a defective protein called the cystic fibrosis transmembrane conductance regulator (CFTR). CFTR is a cyclic AMP-dependent chloride channel found in the membrane of cells. The gene for CFTR is located on chromosome 7. Over 1000 different gene mutations have been discovered that cause a number of distinct defects in CFTR, but by far the most frequent mutation in the UK is delta F508. The correlation between genotype and phenotype is relatively weak for CF lung disease but stronger for gastrointestinal disease. This suggests that additional factors are important in determining the severity of lung disease, including different microbial pathogens, passive smoking, social deprivation and other 'modifier' genes.

Identification of the gene mutation involved within a family allows prenatal diagnosis and carrier detection in the wider family.

Pathophysiology

CF is a multi-system disorder which results mainly from abnormal ion transport across epithelial cells. In the airways this leads to reduction in the airway surface liquid layer and consequent impaired ciliary function and retention of mucopurulent secretions. Chronic endobronchial infection with specific organisms such as *Pseudomonas aeruginosa* ensues. Defective CFTR also causes dysregulation of inflammation and defence against infection. In the intestine, thick viscid meconium is produced, leading to meconium ileus in 10–20% of infants (see below). The pancreatic ducts also become blocked by thick secretions, leading to pancreatic enzyme deficiency and malabsorption. Abnormal function of the sweat glands results in excessive concentrations of sodium and chloride in the sweat.

Clinical features

In the UK, screening of newborns is now performed as part of the heel-prick bloodspot biochemical screen (Guthrie test). The majority of children with CF are identified by screening; however, children may still present clinically with recurrent chest infections, poor growth or malabsorption (Box 16.4). Chronic infection with specific bacteria – initially *Staphylococcus aureus* and *Haemophilus influenzae* and subsequently with *Pseudomonas aeruginosa* or *Burkholderia* species results from viscid mucus in the smaller airways of the lungs. This leads to damage of the bronchial wall, bronchiectasis and abscess formation (Fig. 16.19). The child has a persistent, loose cough, productive of purulent sputum. On examination there is hyperinflation of the chest due to air trapping, coarse inspiratory crepitations and/or expiratory wheeze. With

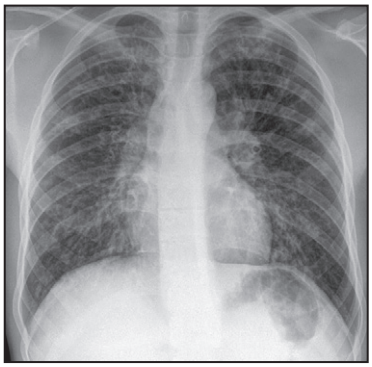


Figure 16.19 A chest X-ray in cystic fibrosis showing hyperinflation, marked peribronchial shadowing, bronchial wall thickening and ring shadows.

Box 16.4 Clinical features of cystic fibrosis

Newborn
- Diagnosed through newborn screening

Infancy
- Meconium ileus in newborn period
- Prolonged neonatal jaundice
- Failure to thrive
- Recurrent chest infections
- Malabsorption, steatorrhoea

Young child
- Bronchiectasis
- Rectal prolapse
- Nasal polyp
- Sinusitis

Older child and adolescent
- Allergic bronchopulmonary aspergillosis (ABPA)
- Diabetes mellitus
- Cirrhosis and portal hypertension
- Distal intestinal obstruction (DIOS, meconium ileus equivalent)
- Pneumothorax or recurrent haemoptysis
- Sterility in males

established disease, there is finger clubbing. Ultimately, 95% die of respiratory failure.

Over 90% of children with CF have pancreatic exocrine insufficiency (lipase, amylase and proteases), resulting in maldigestion and malabsorption. Untreated, this leads to failure to thrive (Fig. 16.20) and passing frequent large, pale, very offensive and greasy stools (steatorrhoea). Pancreatic insufficiency can be diagnosed by demonstrating low elastase in faeces.

About 10–20% of CF infants present in the neonatal period with meconium ileus, in which inspissated meconium causes intestinal obstruction with vomiting, abdominal distension and failure to pass meconium in

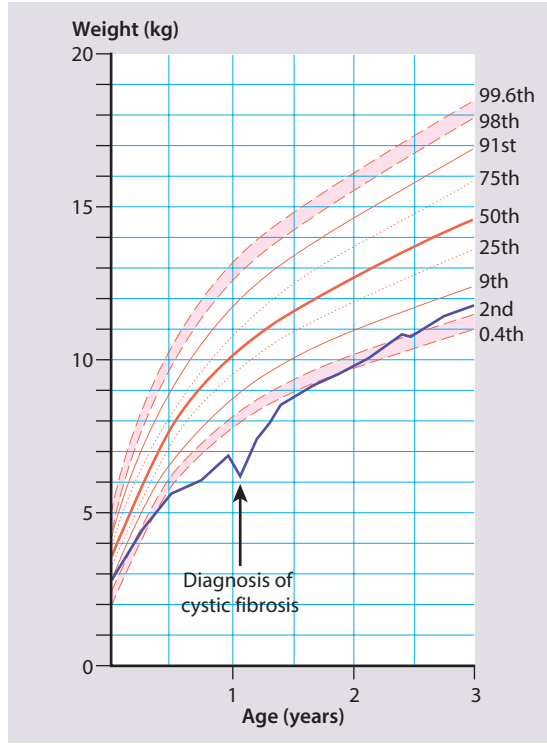


Figure 16.20 Growth chart of a child with cough and recurrent wheeze. Only when the diagnosis of cystic fibrosis was made and appropriate treatment started did he gain weight. (Adapted from growth chart, © RCPCH/WHO/Department of Health.)

the first few days of life. Initial treatment is with Gastrografin enemas, but most cases require surgery.

Diagnosis

The essential diagnostic procedure is the sweat test, to confirm that the concentration of chloride in sweat is markedly elevated (Cl 60–125 mmol/L in cystic fibrosis, 10–40 mmol/L in normal children). Sweating is stimulated by pilocarpine iontophoresis. The sweat is collected into a special capillary tube or absorbed onto a weighed piece of filter paper. Diagnostic errors are common if there is an inadequate volume of sweat collected, so the test must be performed by experienced staff. Confirmation of diagnosis can be made with testing for gene abnormalities in the CFTR protein. If a child is homozygote with two identified mutations then they have cystic fibrosis.

Management

The effective management of CF requires a multidisciplinary team approach, including paediatricians, physiotherapists, dieticians, specialist nurses, the primary care team, teachers and, most importantly, the child and parents. All patients with CF should be reviewed at least annually in a specialist centre. The aims of therapy are to prevent progression of the lung disease and to maintain adequate nutrition and growth.

Respiratory management

Recurrent and persistent bacterial chest infection is the major problem. In younger children, respiratory status is monitored on symptoms; older children should have their lung function measured regularly by spirometry. The FEV_1 (forced expiratory volume in 1 second), expressed as a percentage predicted for age, sex and height, is an indicator of clinical severity and declines with disease progression.

With regular treatment, most infants and children with CF should have no respiratory symptoms, and often have no abnormal signs. From diagnosis, children should have physiotherapy at least twice a day, aiming to clear the airways of secretions. In younger children, parents are taught to perform airway clearance at home using chest percussion and postural drainage. Older patients perform controlled deep breathing exercises and use a variety of physiotherapy devices for airway clearance. Physical exercise is beneficial and is encouraged.

Many CF specialists recommend continuous prophylactic oral antibiotics (usually flucloxacillin), with additional rescue oral antibiotics for any increase in respiratory symptoms or decline in lung function. Persisting symptoms or signs require prompt and vigorous intravenous therapy to limit lung damage, usually administered for 14 days via a peripheral venous long line. Increasingly, parents are taught to administer courses of intravenous antibiotics at home, so decreasing disruption of normal activities such as school. Chronic *Pseudomonas* infection is associated with a more rapid decline in lung function, and this is slowed by the use of daily nebulised antipseudomonal antibiotics. Nebulised DNAse or hypertonic saline may be helpful to decrease the viscosity of sputum and so increase its clearance. The macrolide antibiotic azithromycin, given regularly, decreases respiratory exacerbations, probably due to an immunomodulatory effect rather than antibiotic action. Regular, nebulised hypertonic saline may decrease the number of respiratory exacerbations.

More severe CF requires more regular intravenous antibiotic therapy. If venous access becomes troublesome, implantation of a central venous catheter with a subcutaneous port (e.g. Portacath) simplifies venous access, although they require monthly flushing and complications may develop.

Bilateral sequential lung transplantation is the only therapeutic option for end-stage CF lung disease. Fortunately, this is rarely required during childhood. Outcomes following lung transplantation continue to improve with >50% survival at 10 years. Meticulous assessment, for example with regard to comorbidities and microbiology, psychological preparation, optimal timing of transplantation and expert post-transplant care, are all essential parts of the multidisciplinary transplant process.

Nutritional management

Dietary status should be assessed regularly. Pancreatic insufficiency is treated with oral enteric-coated pancreatic replacement therapy taken with all meals and snacks. Dosage is adjusted according to clinical response. A high-calorie diet is essential, and dietary intake is recommended at 150% of normal. To achieve this, overnight feeding via a gastrostomy is increasingly used. Most patients require fat-soluble vitamin supplements.

Teenagers and adults

Most children with CF now survive into adult life. With increasing age come increased complications, most commonly diabetes mellitus due to decreasing pancreatic endocrine function. Up to one-third of patients will have evidence of liver disease with hepatomegaly on liver palpation, abnormal liver function on blood tests or an abnormal ultrasound; regular ursodeoxycholic acid, to improve flow of bile, may be beneficial. Rarely, the liver disease progresses to cirrhosis, portal hypertension and ultimately liver failure. Liver transplant is generally very successful in CF-related liver failure.

In distal intestinal obstruction syndrome (meconium ileus equivalent), viscid mucofaeculent material obstructs the bowel. This is usually cleared by oral Gastrografin.

There may be increasing chest infections, as well as other late respiratory complications including pneumothorax and life-threatening haemoptysis. There is increasing concern over transmission of virulent strains of *Pseudomonas* and *Burkholderia cepacia* between patients, causing rapid decline in lung function. Consequently, patients are often segregated and advised not to socialise with other people with CF.

Females have normal fertility, and unless they have severe lung disease, tolerate pregnancy well. Males are virtually always infertile due to absence of the vas deferens, although they can father children through intracytoplasmic sperm injection (ICSI).

The psychological repercussions on the affected child and family of a chronic and ultimately fatal illness which requires regular physiotherapy and drugs, frequent hospital admissions and absences from school are considerable. The CF team should provide psychological and emotional support. Adolescents have particular needs which must receive special consideration. Older adolescents with CF should transfer to specialist adult CF care.

Gene therapy is currently being assessed but is unlikely to be of practical value in the immediate future.

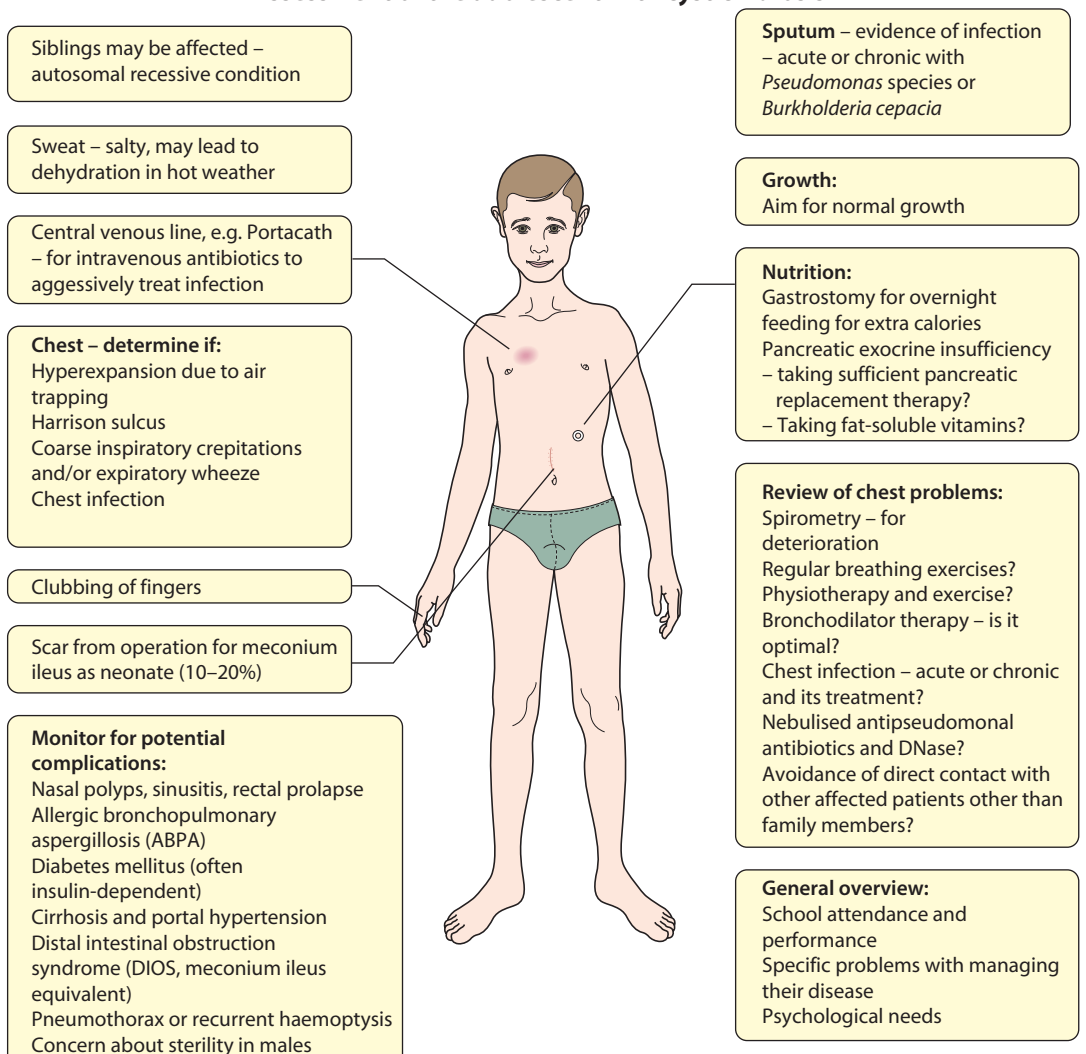
Screening

All newborn infants born in the UK are screened for CF. Immunoreactive trypsinogen (IRT) is raised in CF patients and can be measured in routine heel-prick blood taken for biochemical screening of all babies (Guthrie test). Those samples with a raised IRT are then screened for common CF gene mutations, and infants with two mutations have a sweat test to confirm the diagnosis.

Early identification of CF allows the early introduction of regular treatment. This leads to better nutrition in childhood and improved neurodevelopmental outcome. It also allows proactive institution of

Summary

Assessment of the adolescent with cystic fibrosis



respiratory management and avoids the morbidity and parental anxiety experienced prior to the clinical diagnosis being established. It also enables early genetic counselling for the parents about the one in four risk of recurrence and the possibility of prenatal diagnosis in future pregnancies.

Cystic fibrosis should be considered in any child with recurrent infections, loose stools or failure to thrive.

Sleep-related breathing disorders

This is receiving increased recognition. Up to 12% of pre-pubertal school children snore and estimates of the prevalence of obstructive sleep apnoea (OSA) resulting in gas-exchange abnormalities range from 0.7 to 3%.

Key aspects of the history include loud snoring, witnessed pauses in breathing (apnoeas), restlessness and disturbed sleep. Affected children may be obese, although others may have growth failure. Important consequences of obstructive sleep apnoea include excessive daytime sleepiness, learning and behaviour problems, acute life-threatening cardiorespiratory events and, in severe cases, pulmonary hypertension. In childhood, it is usually due to upper airway obstruction secondary to adenotonsillar hypertrophy. Predisposing causes of sleep-disordered breathing are hypotonia, muscle weakness and anatomical problems, e.g. Down syndrome, achondroplasia, neuromuscular disease, cerebral palsy or craniofacial abnormalities. Such high-risk groups may warrant screening on a regular basis.

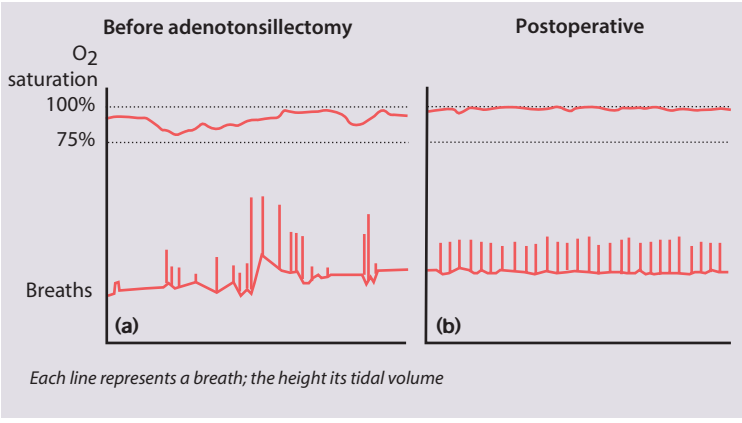


Figure 16.21 Extract from cardiorespiratory monitoring in a child with obstructive sleep apnoea. **(a)** Irregular breathing with periodic pauses associated with oxygen desaturation. **(b)** Post-adenotonsillectomy, the breathing is regular and the desaturation has resolved. (Courtesy of Dr Parviz Habibi.)

The most basic assessment is overnight pulse oximetry, which can be performed in the child's home. The frequency and severity of periods of desaturation (sats <92%) can be quantified. Normal oximetry does not exclude the condition. Limited polysomnography is required in more complex cases; it includes monitoring of heart rate, respiratory effort, airflow, a measure of arterial pCO_2 and video recording. It provides more information about gas exchange and can distinguish between central and obstructive events. Sometimes EEG, electrooculogram and submental EMG is needed to assess neurological arousals and sleep staging.

In cases due to adenotonsillar hypertrophy, adenotonsillectomy is usually curative (Fig. 16.21a,b). Overnight oximetry should be performed prior to surgery for obstructive sleep apnoea to identify severe hypoxaemia, which may increase the risk of perioperative complications. If it persists despite adenotonsillectomy, polysomnography should be performed in a specialist centre. Nasal or facemask continuous positive pressure ventilation (CPAP) or bi-level positive airway pressure (BIPAP) may be required at night.

Congenital central hypoventilation syndrome is a rare congenital condition caused by gene mutations resulting in disordered central control of breathing. In severe cases, life-threatening hypoventilation occurs during sleep, which may result in death in infancy. Long-term ventilation, either continuous or during sleep only, is the mainstay of treatment.

Summary

Sleep disordered breathing

- The majority are due to adenotonsillar hypertrophy, and surgical removal is usually curative

Tracheostomy

The number of children of all ages with a tracheostomy is increasing. Indications are listed in Table 16.3.

Table 16.3 Some indications for tracheostomy in children

Narrow upper airways	Subglottic stenosis
	Laryngeal anomalies (e.g. atresia, haemangiomas, webs)
	Pierre Robin sequence (small jaw and cleft palate)
	Craniofacial anomalies
Lower airway anomalies	Severe tracheo-bronchomalacia
Long-term ventilation	Muscle weakness
	Head or spinal injury
Wean from ventilation	Any prolonged episode of ventilation
Airway protection	Clearance of secretions
	Reduction of aspiration

If a child with a tracheostomy develops sudden and severe breathing difficulties, it may be that the tracheostomy tube is blocked with secretions and needs urgent suction or needs changing immediately. All children with a tracheostomy should have a spare tracheostomy tube with them at all times. If this does not relieve the difficulty in breathing, respiratory support is given via the tracheostomy tube.

Long-term ventilation

An increasing number of children are receiving long-term respiratory support. Preterm infants with severe bronchopulmonary dysplasia (chronic lung disease) may require additional oxygen for many months, and may also require respiratory support with CPAP (continuous positive airway pressure) via nasal prongs or nasal mask. Children with muscle weakness from Duchenne muscular dystrophy, spinal muscular



Figure 16.22 Long-term non-invasive respiratory support given overnight via nasal mask to a child with muscle weakness.



Figure 16.23 Long-term ventilation via a tracheostomy.

atrophy, congenital muscular dystrophy and other rare conditions are increasingly offered long-term ventilatory support. They experience not only hypoxia but also significant hypercapnia due to hypoventilation. This requires bi-level positive airway pressure (BiPAP), which can be delivered non-invasively by a nasal mask or full facemask (Fig. 16.22). In some cases BiPAP may need to be delivered via a tracheostomy (Fig. 16.23). In Duchenne muscular dystrophy and some other conditions causing muscle weakness, non-invasive ventilation at night provides additional quality years of life. This service can often be provided at home, with considerable specialist community support. With progressive neurological disorders, difficult ethical decisions need to be made about admission for intensive care and initiation of long-term full ventilation.

Further reading

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Definitive textbook.

17

Cardiac disorders

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Recent developments in paediatric cardiac disease are:

- Lesions are being increasingly identified on antenatal ultrasound screening
- Most lesions are diagnosed by echocardiography, the mainstay of diagnostic imaging
- MRI allows three-dimensional reconstruction of complex cardiac disorders, assessment of haemodynamics and flow patterns and assists interventional cardiology, reducing the need for cardiac catheterisation
- Most defects can be corrected by definitive surgery at the initial operation
- An increasing number of defects (60%) are treated non-invasively, e.g. persistent ductus arteriosus
- New therapies are available to treat pulmonary hypertension and delay transplantation
- The overall infant cardiac surgical mortality has been reduced from approximately 20% in 1970 to 2% in 2010.

Epidemiology

Heart disease in children is mostly congenital. It is the most common single group of structural malformations in infants:

- 8 per 1000 liveborn infants have significant cardiac malformations
- Some abnormality of the cardiovascular system, e.g. a bicuspid aortic valve, is present in 1–2% of live births
- About 1 in 10 stillborn infants have a cardiac anomaly.

The nine most common anomalies account for 80% of all lesions (Box 17.1), but:

- about 10–15% have complex lesions with more than one cardiac abnormality and
- about 10–15% also have a non-cardiac abnormality.

Box 17.1 The most common congenital heart lesions

Left-to-right shunts (breathless)

- Ventricular septal defect 30%
- Persistent arterial duct 12%
- Atrial septal defect 7%

Right-to-left shunts (blue)

- Tetralogy of Fallot 5%
- Transposition of the great arteries 5%

Common mixing (breathless and blue)

- Atrioventricular septal defect (complete) 2%

Outflow obstruction in a well child (asymptomatic with a murmur)

- Pulmonary stenosis 7%
- Aortic stenosis 5%

Outflow obstruction in a sick neonate (collapsed with shock)

- Coarctation of the aorta 5%.

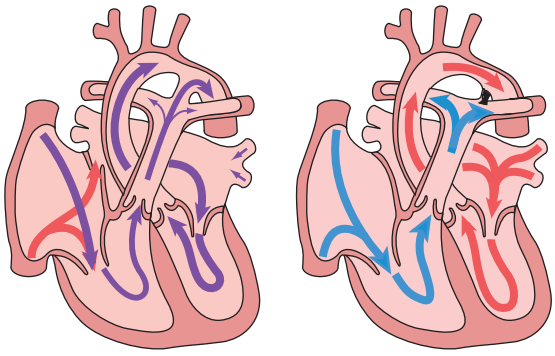
Aetiology

Genetic causes are increasingly recognised in the aetiology of congenital heart disease, now in more than 10%. These might affect whole chromosomes, point mutations or microdeletions (Table 17.1). Polygenic abnormalities probably explain why having a child with congenital heart disease doubles the risk for subsequent children and the risk is higher still if either parent has congenital heart disease. A small proportion are related to external teratogens.

Circulatory changes at birth

In the fetus, the left atrial pressure is low, as relatively little blood returns from the lungs. The pressure in the right atrium is higher than in the left, as it receives all the systemic venous return including blood from the placenta. The flap valve of the foramen ovale is held open, blood flows across the atrial septum into the left atrium and then into the left ventricle, which in turn pumps it to the upper body (Fig. 17.1).

With the first breaths, resistance to pulmonary blood flow falls and the volume of blood flowing through the lungs increases six-fold. This results in a rise in the left atrial pressure. Meanwhile, the volume of blood returning to the right atrium falls as the placenta is excluded from the circulation. The change in the pressure difference causes the flap valve of the foramen ovale to be closed. The ductus arteriosus, which connects the pulmonary artery to the aorta in fetal life, will normally close within the first few hours or days. Some babies with congenital heart lesions rely on blood flow through the duct (duct-dependent circulation). Their clinical condition will deteriorate dramatically when the duct closes,



Antenatal circulation Postnatal circulation

Figure 17.1 Changes in the circulation from the fetus to the newborn. When congenital heart lesions rely on blood flow through the duct (a duct-dependent circulation), there will be a dramatic deterioration in the clinical condition when the duct closes.

Table 17.1 Causes of congenital heart disease

	Cardiac abnormalities	Frequency
Maternal disorders		
Rubella infection	Peripheral pulmonary stenosis, PDA	30–35%
Systemic lupus erythematosus (SLE)	Complete heart block (anti-Ro and anti-La antibody)	35%
Diabetes mellitus	Incidence increased overall	2%
Maternal drugs		
Warfarin therapy	Pulmonary valve stenosis, PDA	5%
Fetal alcohol syndrome	ASD, VSD, tetralogy of Fallot	25%
Chromosomal abnormality		
Down syndrome (trisomy 21)	Atrioventricular septal defect, VSD	30%
Edwards syndrome (trisomy 18)	Complex	60–80%
Patau syndrome (trisomy 13)	Complex	70%
Turner syndrome (45XO)	Aortic valve stenosis, coarctation of the aorta	15%
Chromosome 22q11.2 deletion	Aortic arch anomalies, tetralogy of Fallot, common arterial trunk	80%
Williams syndrome (7q11.23 microdeletion)	Supravalvular aortic stenosis, peripheral pulmonary artery stenosis	85%
Noonan syndrome (PTPN11 mutation and others)	Hypertrophic cardiomyopathy, atrial septal defect, pulmonary valve stenosis	50%

ASD, atrial septal defect; PDA, persistent ductus arteriosus; VSD, ventricular septal defect.

which is usually at 1–2 days of age but occasionally later.

Presentation

Congenital heart disease presents with:

- Antenatal cardiac ultrasound diagnosis
- Detection of a heart murmur
- Heart failure
- Shock.
- Cyanosis.

Antenatal diagnosis

Checking the anatomy of the fetal heart has become a routine part of the fetal anomaly scan performed in developed countries between 18 and 20 weeks' gestation and can lead to 70% of those infants who require surgery in the first 6 months of life being diagnosed antenatally. If an abnormality is detected, detailed fetal echocardiography is performed by a paediatric cardiologist. Any fetus at increased risk, e.g. suspected Down syndrome, where the parents have had a previous child with heart disease or where the mother has congenital heart disease, is also checked. Early diagnosis allows the parents to be counselled. Depending on the diagnosis, some choose termination of pregnancy; the majority continue with the pregnancy and can have their child's management planned antenatally. Mothers of infants with duct-dependent lesions likely to need treatment within the first 2 days of life may be offered delivery at or close to the cardiac centre.

Heart murmurs

The most common presentation of congenital heart disease is with a heart murmur. Even so, the vast majority of children with murmurs have a normal heart. They have an 'innocent murmur', which can be heard at some time in almost 30% of children. It is obviously important to be able to distinguish an innocent murmur from a pathological one.

Hallmarks of an innocent ejection murmur are (all have an 'S', 'innoSent'):

- aSymptomatic patient
- Soft blowing murmur
- Systolic murmur only, not diastolic
- left Sternal edge.

Also:

- Normal heart sounds with no added sounds
- No parasternal thrill
- No radiation.

During a febrile illness or anaemia, innocent or flow murmurs are often heard because of increased cardiac output. Therefore it is important to examine the child when such other illnesses have been corrected.

Differentiating between innocent and pathological murmurs can be difficult. If a murmur is thought to be significant, or if there is uncertainty about whether it is innocent, the child should be seen by an experienced paediatrician to decide about referral to a paediatric cardiologist for echocardiography. A chest radiograph and ECG may help with the diagnosis beyond the neonatal period.

Many newborn infants with potential shunts have neither symptoms nor a murmur at birth, as the pulmonary vascular resistance is still high. Therefore, conditions such as a ventricular septal defect or ductus arteriosus may only become apparent at several weeks of age when the pulmonary vascular resistance falls.

The features of an innocent murmur can be remembered as the five Ss: 'InnoSent' murmur = Soft, Systolic, aSymptomatic, left Sternal edge.

Heart failure

Symptoms

- Breathlessness (particularly on feeding or exertion)
- Sweating
- Poor feeding
- Recurrent chest infections.

Signs

- Poor weight gain or 'faltering growth'
- Tachypnoea
- Tachycardia
- Heart murmur, gallop rhythm
- Enlarged heart
- Hepatomegaly
- Cool peripheries.

Signs of right heart failure (ankle oedema, sacral oedema and ascites) are rare in developed countries, but may be seen with long-standing rheumatic fever or pulmonary hypertension, with tricuspid regurgitation and right atrial dilatation.

In the first week of life, heart failure (Box 17.2) usually results from left heart obstruction, e.g. coarctation of the aorta. If the obstructive lesion is very severe then

Box 17.2 Causes of heart failure

1. Neonates – obstructed (duct-dependent) systemic circulation

- Hypoplastic left heart syndrome
- Critical aortic valve stenosis
- Severe coarctation of the aorta
- Interruption of the aortic arch

2. Infants (high pulmonary blood flow)

- Ventricular septal defect
- Atrioventricular septal defect
- Large persistent ductus arteriosus

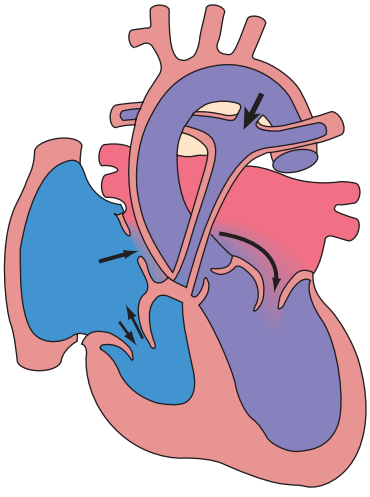
3. Older children and adolescents (right or left heart failure)

- Eisenmenger syndrome (right heart failure only)
- Rheumatic heart disease
- Cardiomyopathy.

arterial perfusion may be predominantly by right-to-left flow of blood via the arterial duct, so-called duct-dependent systemic circulation (Fig. 17.2). Closure of the duct under these circumstances rapidly leads to severe acidosis, collapse and death unless ductal patency is restored (Case History 17.1).

After the first week of life, progressive heart failure is most likely due to a left-to-right shunt (Case History 17.2). During the subsequent weeks, as the pulmonary vascular resistance falls, there is a progressive increase in left-to-right shunt and increasing pulmonary blood flow. This causes pulmonary oedema and breathlessness.

Such symptoms of heart failure will increase up to the age of about 3 months, but may subsequently improve as the pulmonary vascular resistance rises in response to the left-to-right shunt. If left untreated, these children will develop Eisenmenger syndrome, which is irreversibly raised pulmonary vascular resistance resulting from chronically raised pulmonary arterial pressure and flow. Now the shunt is from right to left and the teenager is blue. If this develops, the only surgical option is a heart-lung transplant, if available, although medication is now available to palliate the symptoms.



Pulmonary atresia with intact septum

Figure 17.2 The pulmonary circulation is maintained by blood flowing left to right across the duct – an example of a duct-dependent pulmonary circulation.

Cyanosis

- Peripheral cyanosis (blueness of the hands and feet) may occur when a child is cold or unwell from any cause or with polycythaemia
- Central cyanosis, seen on the tongue as a slate blue colour, is associated with a fall in arterial blood oxygen tension. It can only be recognised clinically if the concentration of reduced haemoglobin in the blood exceeds 5 g/dl, so it is less pronounced if the child is anaemic
- Check with a pulse oximeter that an infant's oxygen saturation is normal (≥94%). Persistent cyanosis in an otherwise well infant is nearly always a sign of structural heart disease.

Cyanosis in a newborn infant with respiratory distress (respiratory rate >60 breaths/min) may be due to:

- Cardiac disorders – cyanotic congenital heart disease
- Respiratory disorders, e.g. surfactant deficiency, meconium aspiration, pulmonary hypoplasia, etc.
- Persistent pulmonary hypertension of the newborn (PPHN) – failure of the pulmonary vascular resistance to fall after birth
- Infection – septicaemia from group B streptococcus and other organisms
- Metabolic disease – metabolic acidosis and shock.

Whether the presentation of congenital heart disease is with a heart murmur, heart failure, cyanosis or shock depends on the underlying anatomic lesion causing:

- left to right shunt
- right to left shunt
- common mixing
- outflow obstruction in the well or sick child.

This is summarised in Table 17.2.

Table 17.2 Types of presentation with congenital heart disease

Type of lesion	Left-to-right shunt	Right-to-left shunt	Common mixing	Well children with obstruction	Sick neonates with obstruction
Symptoms	Breathless or asymptomatic	Blue	Breathless and blue	Asymptomatic	Collapsed with shock
Examples	ASD	Tetralogy of Fallot	AVSD	AS	Coarctation
	VSD PDA	TGA	Complex congenital heart disease	PS Adult-type CoA	HLHS

ASD, atrial septal defect; VSD, ventricular septal defect; PDA, patent ductus arteriosus; TGA, transposition of the great arteries; AVSD, atrioventricular; AS, aortic stenosis; PS, pulmonary stenosis; CoA, coarctation of the aorta; HLHS, hypoplastic left heart syndrome.

Case History

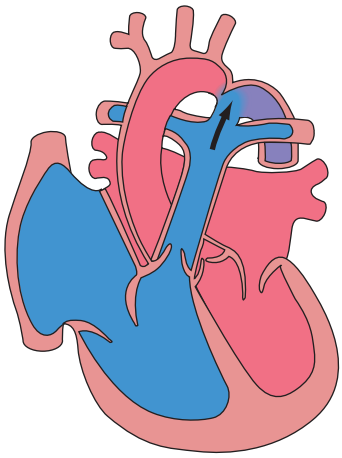
17.1 Shock

A 2-day-old baby had been discharged home the day after delivery following a normal routine examination. He suddenly collapsed and was rushed to hospital. He was pale, with grey lips. The right brachial pulse could just be felt, the femoral pulses were impalpable and his liver was significantly enlarged. Blood gases showed a severe metabolic acidosis. The differential diagnosis was:

- Congenital heart disease
- Septicaemia
- Inherited disorder of metabolism.

He was ventilated and treated with volume support. Blood cultures were taken and antibiotics started for possible sepsis. Blood and urine samples were taken for an amino acid screen and urine for organic acids. As the femoral pulses remained impalpable, a prostaglandin infusion was started. Within 2 hours, he was pink and well perfused and the acidosis was resolving. Severe coarctation of the aorta (Fig. 17.3) was diagnosed on echocardiography. He had developed shock from a left heart outflow tract obstruction once the arterial duct had closed.

Maintaining ductal patency is the key to early survival in neonates with a duct-dependent circulation.



Duct-dependent coarctation

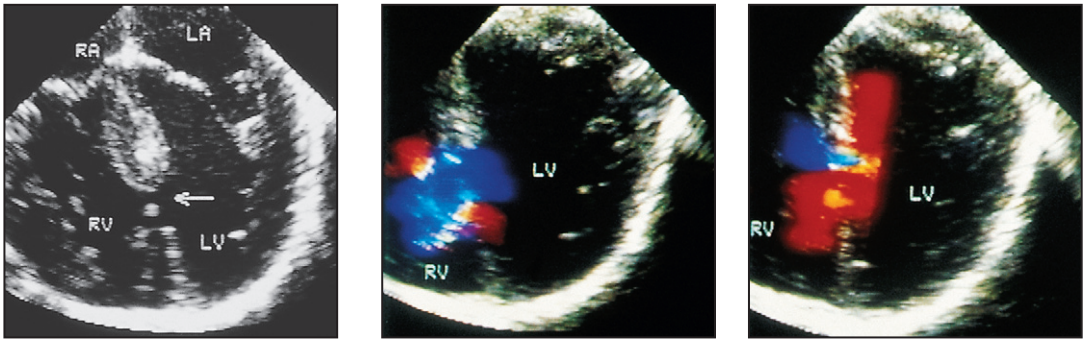
Figure 17.3 The systemic circulation is maintained by blood flowing right to left across the ductus arteriosus – a duct-dependent systemic circulation.

Case History

17.2 Heart failure

A 5-week-old female infant was referred to hospital because of wheezing, poor feeding and poor weight gain during the previous 2 weeks. Before this, she had been well. Her routine neonatal examination had been normal. She was tachypnoeic (50–60 breaths/min) and there was some sternal and intercostal recession. The pulses were normal. There was a thrill, a pansystolic murmur at the lower left sternal edge and a slightly accentuated pulmonary component to the second heart sound. There were scattered wheezes. The liver was enlarged, palpable at two fingerbreadths below the costal margin. The ECG was unremarkable. The chest radiograph showed cardiomegaly and increased pulmonary vascular markings. An echocardiogram showed a moderate-sized ventricular septal defect (VSD) (Fig. 17.4). Treatment was medical with diuretics and captopril. The VSD closed spontaneously at 18 months.

This infant developed heart failure from a moderate VSD presenting at several weeks of age when the pulmonary resistance fell, causing increased left-to-right shunting of blood. The defect closed spontaneously.



(a) (b) (c)

Figure 17.4 **(a)** Echocardiogram showing a medium-sized muscular ventricular septal defect (arrow). **(b)** The colour Doppler shows a left-to-right shunt (blue) during systole. **(c)** There is also a small right-to-left shunt (red) during diastole (RA, right atrium; LA, left atrium; RV, right ventricle; LV, left ventricle).

Summary

Presentation of congenital heart disease

- Antenatal ultrasound screening – increasing proportion detected
- Detection of a heart murmur – need to differentiate innocent from pathological murmur
- Cyanosis – if duct dependent, prostaglandin to maintain ductal patency is vital for initial survival
- Heart failure – usually from left-to-right shunt when pulmonary vascular resistance falls
- Shock – when duct closes in severe left heart obstruction.

Diagnosis

If congenital heart disease is suspected, a chest radiograph and ECG (Box 17.3) should be performed. Although rarely diagnostic, they may be helpful in establishing that there is an abnormality of the cardiovascular system and as a baseline for assessing future changes. Echocardiography, combined with Doppler ultrasound, enables almost all causes of congenital heart disease to be diagnosed. Even when a paediatric cardiologist is not available locally a specialist echocardiography opinion may be available via telemedicine, or else transfer to the cardiac centre will be necessary. A specialist opinion is required if the child is haemodynamically unstable, if there is heart failure, if there is cyanosis, when the oxygen saturations are <94% due to heart disease and when there are reduced volume pulses.

Nomenclature

The European (as opposed to American) system for naming congenital heart disease is referred to as sequential segmental arrangement. The advantage is that it is not necessary to remember the pattern of an eponymous syndrome, e.g. tetralogy of Fallot. The disadvantage is that it is longwinded. The idea is that each component is described in turn, naming the way the atria, then the ventricles and then the great arteries are connected. Hence, a normal heart will be described as situs solitus (i.e. the atria are in the correct orientation), concordant atrioventricular connection and concordant ventriculo–arterial connection. Therefore a heart of any complexity can be described in a logical step-by-step process. This system is not described here, as it is beyond the scope of this book.

Box 17.3 ECGs in children

Important features

- Arrhythmias
- Superior QRS axis (negative deflection in AVF) (see Fig. 17.5f)
- Right ventricular hypertrophy (upright T wave in V_1, over 1 month of age) (see Fig. 17.6e)
- Left ventricular strain (inverted T wave in V_6) (see Fig. 17.13d)

Pitfalls

- P-wave morphology is rarely helpful in children
- Partial right bundle branch block – most are normal children, although it is common in ASD
- Sinus arrhythmia is a normal finding.

ASD, atrial septal defect.

Left-to-right shunts

These are:

- Atrial septal defects
- Ventricular septal defects
- Persistent ductus arteriosus.

Atrial septal defect

There are two main types of atrial septal defect (ASD):

- Secundum ASD (80% of ASDs) (Fig. 17.5a)
- Partial atrioventricular septal defect (primum ASD, pAVSD) (Fig 17.5b).

Both present with similar symptoms and signs, but their anatomy is quite different. The secundum ASD is a defect in the centre of the atrial septum involving the foramen ovale.

Partial AVSD is a defect of the atrioventricular septum and is characterised by:

- An inter-atrial communication between the bottom end of the atrial septum and the atrioventricular valves (primum ASD)
- Abnormal atrioventricular valves, with a left atrioventricular valve which has three leaflets and tends to leak (regurgitant valve).

Clinical features

Symptoms

- None (commonly)
- Recurrent chest infections/wheeze
- Arrhythmias (fourth decade onwards).

Physical signs (Fig. 17.5c)

- An ejection systolic murmur best heard at the upper left sternal edge – due to increased flow across the pulmonary valve because of the left-to-right shunt
- A fixed and widely split second heart sound (often difficult to hear) – due to the right ventricular stroke volume being equal in both inspiration and expiration
- With a partial AVSD, an apical pansystolic murmur from atrioventricular valve regurgitation.

Investigations

Chest radiograph (Fig. 17.5d)

Cardiomegaly, enlarged pulmonary arteries and increased pulmonary vascular markings.

Atrial septal defect

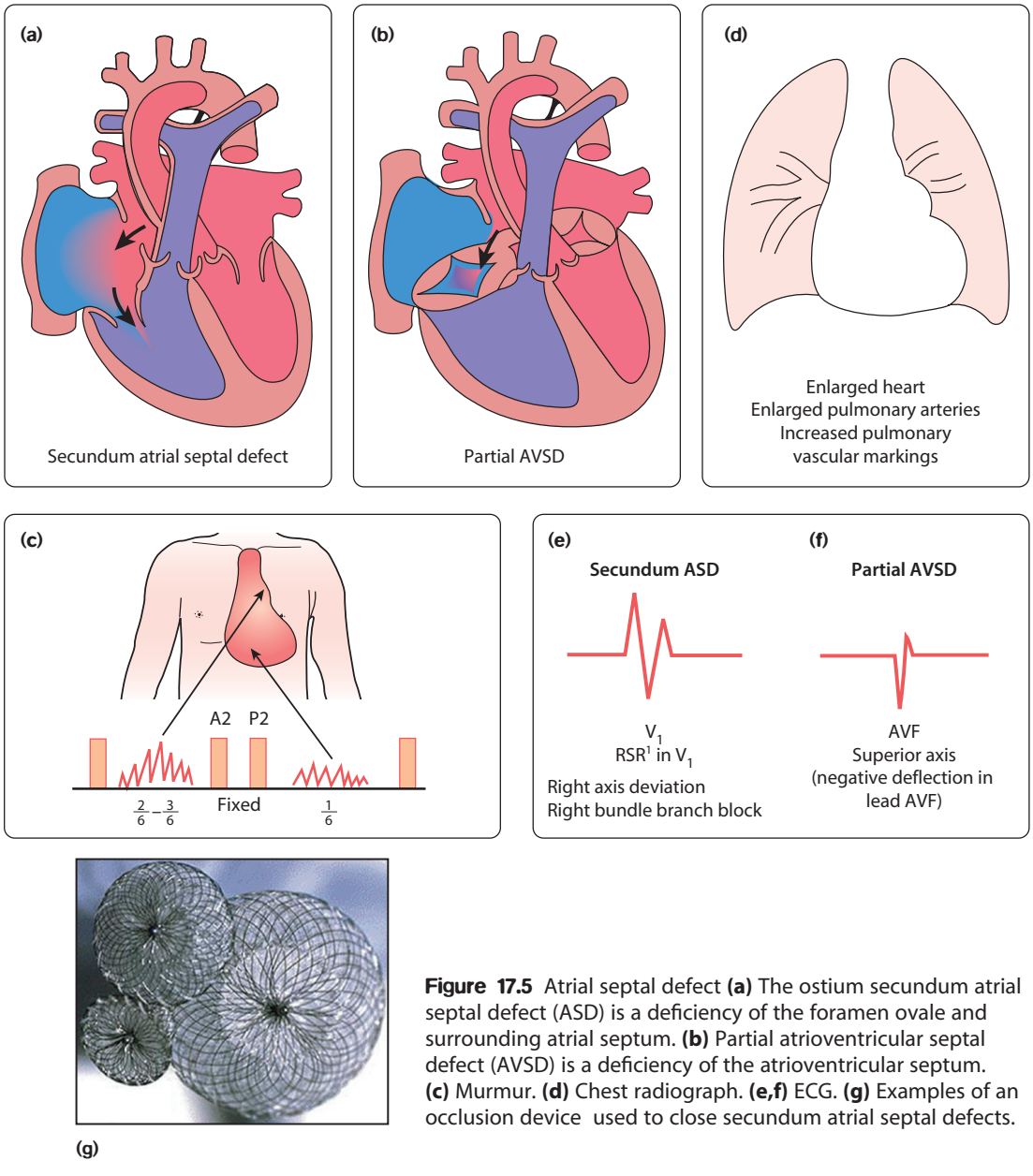


Figure 17.5 Atrial septal defect **(a)** The ostium secundum atrial septal defect (ASD) is a deficiency of the foramen ovale and surrounding atrial septum. **(b)** Partial atrioventricular septal defect (AVSD) is a deficiency of the atrioventricular septum. **(c)** Murmur. **(d)** Chest radiograph. **(e,f)** ECG. **(g)** Examples of an occlusion device used to close secundum atrial septal defects.

ECG

- Secundum ASD – partial right bundle branch block is common (but may occur in normal children), right axis deviation due to right ventricular enlargement (Fig. 17.5e)
- Partial AVSD – a 'superior' QRS axis (mainly negative in AVF) (Fig 17.5f). This occurs because there is a defect of the middle part of the heart where the atrioventricular node is. The displaced node then conducts to the ventricles superiorly, giving the abnormal axis.

Echocardiography

This will delineate the anatomy and is the mainstay of diagnostic investigations.

Management

Children with significant atrial septal defect (large enough to cause right ventricle dilation) will require treatment. For secundum ASDs, this is by cardiac catheterisation with insertion of an occlusion device (Fig 17.5g), but for partial AVSD, surgical correction is

required. Treatment is usually undertaken at about 3–5 years of age in order to prevent right heart failure and arrhythmias in later life.

Ventricular septal defects

Ventricular septal defects (VSDs) are common, accounting for 30% of all cases of congenital heart disease. There is a defect anywhere in the ventricular septum, perimembranous (adjacent to the tricuspid valve) or muscular (completely surrounded by muscle). They can most conveniently be considered according to the size of the lesion.

Small VSDs

These are smaller than the aortic valve in diameter, perhaps up to 3 mm.

Clinical features

Symptoms

- Asymptomatic.

Physical signs

- Loud pansystolic murmur at lower left sternal edge (loud murmur implies smaller defect)
- Quiet pulmonary second sound (P_2).

Investigations

Chest radiograph

- Normal.

ECG

- Normal.

Echocardiography

- Demonstrates the precise anatomy of the defect. It is possible to assess its haemodynamic effects using Doppler echocardiography. There is no pulmonary hypertension.

Management

These lesions will close spontaneously. This is ascertained by the disappearance of the murmur with a normal ECG on follow-up by a paediatrician or paediatric cardiologist and by a normal echocardiogram. While the VSD is present, prevention of bacterial endocarditis is by maintaining good dental hygiene.

Large VSDs

These defects are the same size or bigger than the aortic valve (Fig. 17.6a).

Clinical features

Symptoms

- Heart failure with breathlessness and failure to thrive (faltering growth) after 1 week old
- Recurrent chest infections.

Physical signs (Fig. 17.6b)

- Tachypnoea, tachycardia and enlarged liver from heart failure
- Active precordium
- Soft pansystolic murmur or no murmur (implying large defect)
- Apical mid-diastolic murmur (from increased flow across the mitral valve after the blood has circulated through the lungs)

Summary

Left-to-right shunts

Lesion	Symptoms	Signs	Management
ASD			
Secundum	None	ESM at ULSE Fixed split S_2	Catheter device closure at 3–5 years
Partial AVSD	None	ESM at ULSE Fixed split S_2 Pansystolic murmur at apex	Surgery at 3 years
VSD			
Small (80–90% of cases)	None	Pansystolic murmur at LLSE	None
Large (10–20% if cases)	Heart failure	Active precordium, loud P_2, soft murmur, tachypnoea, hepatomegaly	Diuretics, captopril, calories Surgery at 3–6 months old
PDA	None	Continuous murmur at ULSE ± bounding pulses	Coil or device closure at cardiac catheter at 1 year, or ligation

ASD, atrial septal defect; AVSD, atrioventricular septal defect; VSD, ventricular septal defect; PDA, persistent ductus arteriosus; ESM, ejection systolic murmur; ULSE, upper left sternal edge; LLSE, lower left sternal edge.

Large ventricular septal defect

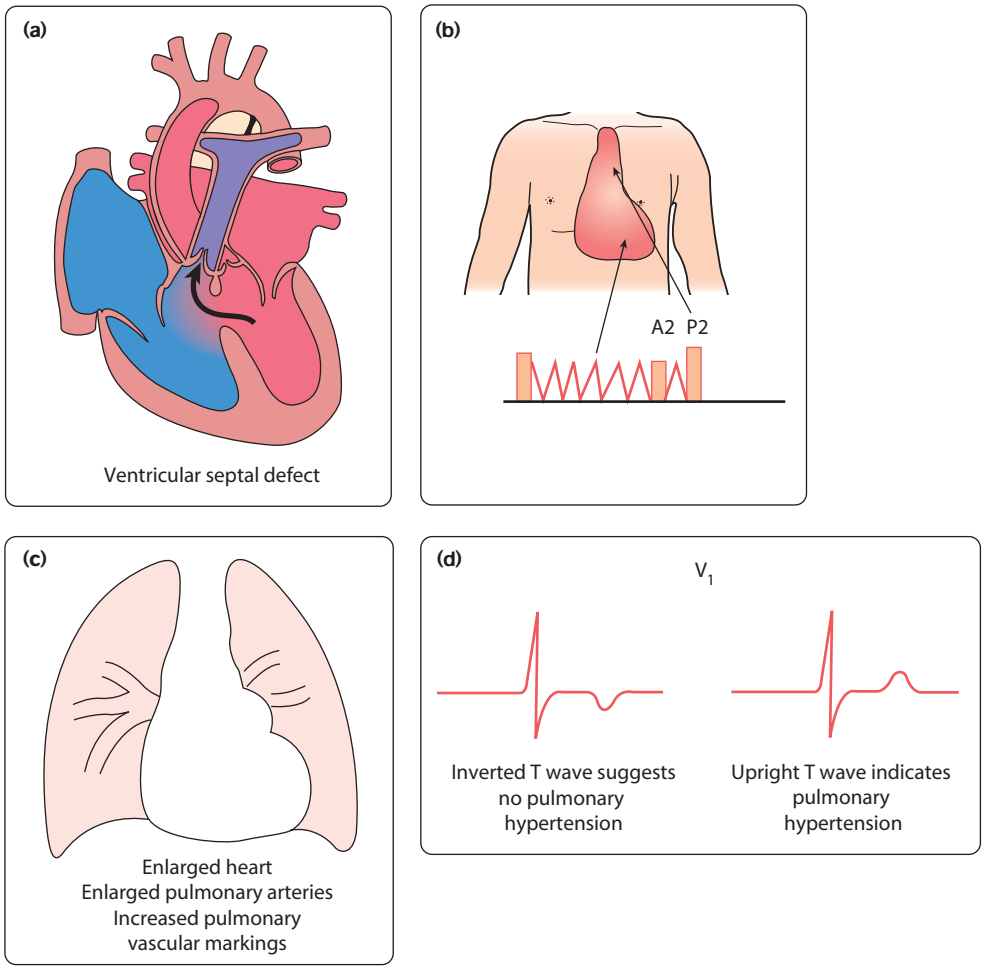


Figure 17.6 Ventricular septal defect. **(a)** Ventricular septal defect showing a left-to-right shunt. **(b)** Murmur. **(c)** Chest radiograph. **(d)** ECG.

- Loud pulmonary second sound (P_2) – from raised pulmonary arterial pressure.

Investigations

Chest radiograph (Fig. 17.6c)

- Cardiomegaly
- Enlarged pulmonary arteries
- Increased pulmonary vascular markings
- Pulmonary oedema.

ECG (Fig. 17.6d)

- Biventricular hypertrophy by 2 months of age.

Echocardiography

- Demonstrates the anatomy of the defect, haemodynamic effects and pulmonary hypertension (due to high flow).

Management

Drug therapy for heart failure is with diuretics, often combined with captopril. Additional calorie input is required. There is always pulmonary hypertension in children with large VSD and left-to-right shunt. This will ultimately lead to irreversible damage of the pulmonary capillary vascular bed (see Eisenmenger syndrome, below). To prevent this, surgery is usually performed at 3–6 months of age in order to:

- Manage heart failure and failure to thrive
- Prevent permanent lung damage from pulmonary hypertension and high blood flow.

Persistent ductus arteriosus (PDA, persistent arterial duct)

The ductus arteriosus connects the pulmonary artery to the descending aorta. In term infants, it normally closes shortly after birth. In persistent ductus arteriosus it has failed to close by 1 month after the expected date of delivery due to a defect in the constrictor mechanism of the duct. The flow of blood across a persistent ductus arteriosus (PDA) is then from the aorta to the pulmonary artery (i.e. left to right), following the fall in pulmonary vascular resistance after birth. In the preterm infant, the presence of a persistent ductus

Persistent ductus arteriosus

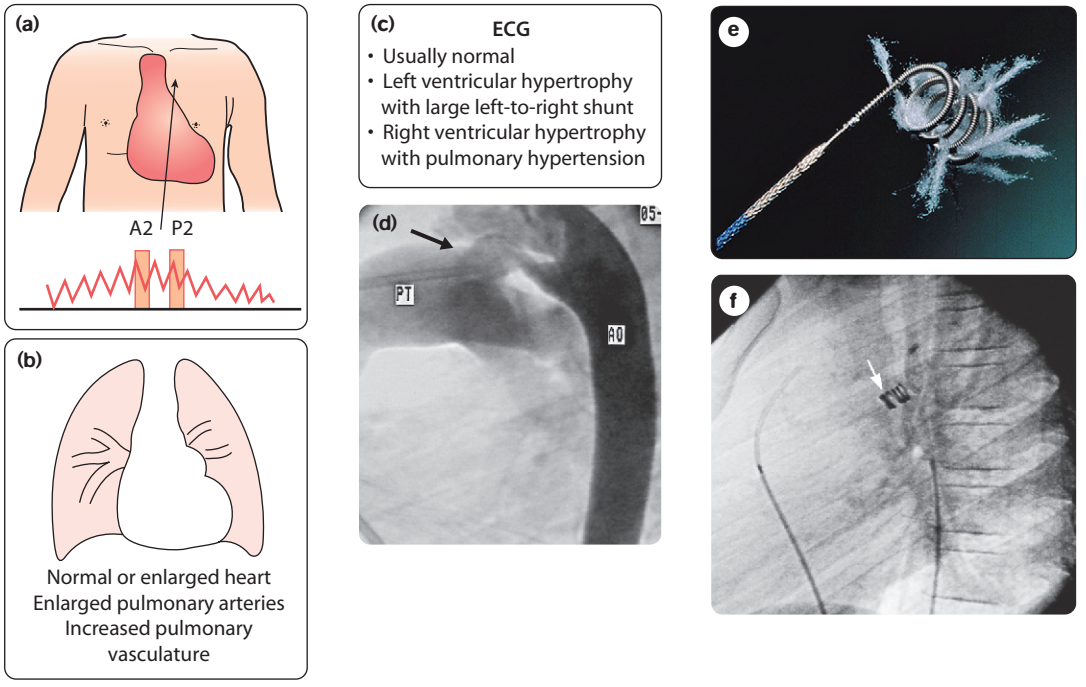


Figure 17.7 Persistent ductus arteriosus. **(a)** Murmur. **(b)** Chest radiograph. **(c)** ECG. **(d)** A persistent ductus arteriosus visualised on angiography. **(e)** A coil used to close ducts. It is passed through a catheter via the femoral artery or vein. **(f)** Angiogram to show coil in the duct. (PT, pulmonary trunk; AO, aorta.)

arteriosus is not from congenital heart disease but due to prematurity. This is described in Chapter 9.

Clinical features

Most children present with a continuous murmur beneath the left clavicle (Fig. 17.7a). The murmur continues into diastole because the pressure in the pulmonary artery is lower than that in the aorta throughout the cardiac cycle. The pulse pressure is increased, causing a collapsing or bounding pulse. Symptoms are unusual, but when the duct is large there will be increased pulmonary blood flow with heart failure and pulmonary hypertension.

Investigations

The chest radiograph and ECG are usually normal, but if the PDA is large and symptomatic the features on chest radiograph (Fig. 17.7b) and ECG (Fig. 17.7c) are indistinguishable from those seen in a patient with a large VSD. However, the duct is readily identified on echocardiography.

Management

Closure is recommended to abolish the lifelong risk of bacterial endocarditis and of pulmonary vascular disease. Closure is with a coil or occlusion device introduced via a cardiac catheter at about 1 year of age (Fig. 17.7d–f). Occasionally, surgical ligation is required.

Right-to-left shunts

These are:

- Tetralogy of Fallot
- Transposition of the great arteries.

Presentation is with cyanosis (blue, oxygen saturations ≤94% or collapsed), usually in the first week of life.

Hyperoxia (nitrogen washout) test

The test is used to help determine the presence of heart disease in a cyanosed neonate. The infant is placed in 100% oxygen (headbox or ventilator) for 10 min. If the right radial arterial PaO_2 from a blood gas remains low (<15 kPa, 113 mmHg) after this time, a diagnosis of 'cyanotic' congenital heart disease can be made if lung disease and persistent pulmonary hypertension of the newborn (persistent fetal circulation) have been excluded. If the PaO_2 is >20 kPa, it is not cyanotic heart disease. Blood gas analysis must be performed as oxygen saturations are not reliable enough in this range of values.

Management

- Stabilise the airway, breathing and circulation (ABC), with artificial ventilation if necessary

- Start prostaglandin infusion (PGE, 5 ng/kg per min). Most infants with cyanotic heart disease presenting in the first few days of life are duct dependent; i.e. there is reduced mixing between the pink oxygenated blood returning from the lungs and the blue deoxygenated blood from the body. Maintenance of ductal patency is the key to early survival of these children. Observe for potential side-effects – apnoea, jitteriness and seizures, flushing, vasodilatation and hypotension.

Tetralogy of Fallot

This is the most common cause of cyanotic congenital heart disease (Fig. 17.8a).

Clinical features

In tetralogy of Fallot, as implied by the name, there are four cardinal anatomical features:

- A large VSD
- Overriding of the aorta with respect to the ventricular septum
- Subpulmonary stenosis causing right ventricular outflow tract obstruction
- Right ventricular hypertrophy as a result.

Symptoms

Most are diagnosed:

- antenatally *or*
- following the identification of a murmur in the first 2 months of life. Cyanosis at this stage may not be obvious, although a few present with severe cyanosis in the first few days of life.

The classical description of severe cyanosis, hypercyanotic spells and squatting on exercise, developing in late infancy, is now rare in developed countries, but still common where access to the necessary paediatric cardiac services is not available. However, it is important to recognise hypercyanotic spells, as they may lead to myocardial infarction, cerebrovascular accidents and even death if left untreated. They are characterised by a rapid increase in cyanosis, usually associated with irritability or inconsolable crying because of severe hypoxia, and breathlessness and pallor because of tissue acidosis. On auscultation, there is a very short murmur during a spell.

Signs

- Clubbing of the fingers and toes will develop in older children
- A loud harsh ejection systolic murmur at the left sternal edge from day 1 of life (Fig. 17.8b). With

Tetralogy of Fallot

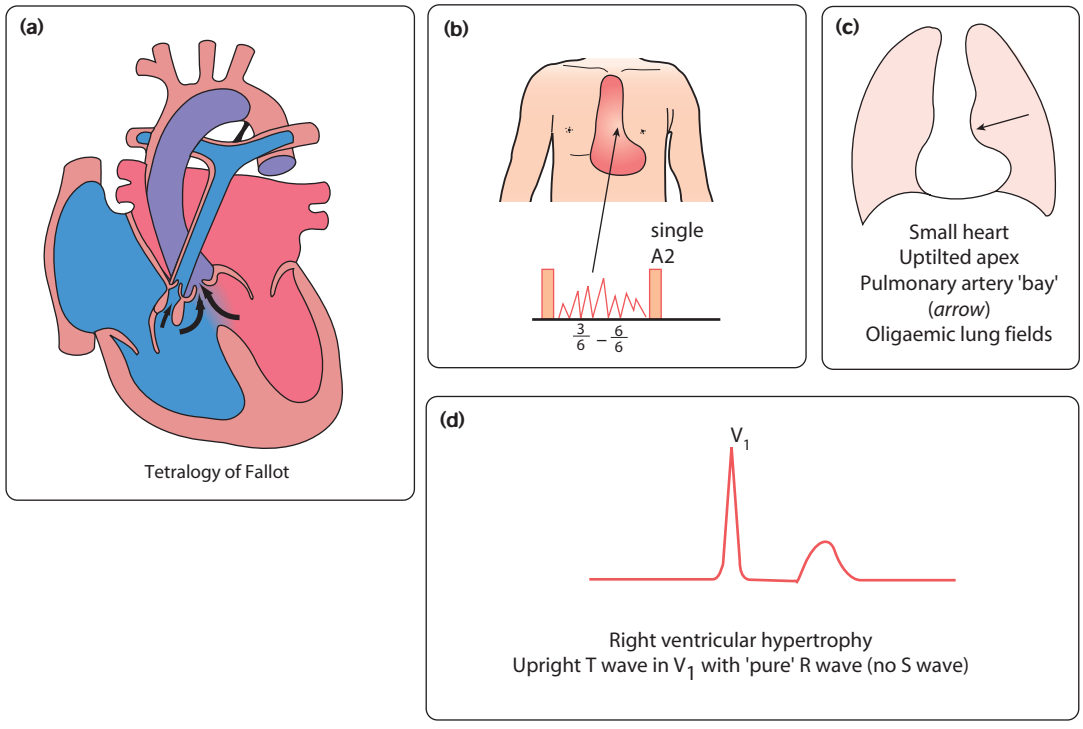


Figure 17.8 Tetralogy of Fallot. **(a)** Tetralogy of Fallot. The right ventricular outflow tract obstruction results in blood flowing from right to left across the ventricular septal defect **(b)** Murmur. **(c)** Chest radiograph. **(d)** ECG.

increasing right ventricular outflow tract obstruction, which is predominantly muscular and below the pulmonary valve, the murmur will shorten and cyanosis will increase.

Investigations

Chest radiograph (Fig. 17.8c)

A radiograph will show a relatively small heart, possibly with an uptilted apex (boot-shaped) due to right ventricular hypertrophy, more prominent in the older child. There may be a right-sided aortic arch, but characteristically, there is a pulmonary artery 'bay', a concavity on the left heart border where the convex-shaped main pulmonary artery and right ventricular outflow tract would normally be profiled. There may also be decreased pulmonary vascular markings reflecting reduced pulmonary blood flow.

ECG (Fig. 17.8d)

Normal at birth. Right ventricular hypertrophy when older.

Echocardiography

This will demonstrate the cardinal features, but cardiac catheterisation may be required to show the detailed anatomy of the coronary arteries.

Management

- Initial management is medical, with definitive surgery at around 6 months of age. It involves closing the VSD and relieving right ventricular outflow tract obstruction, sometimes with an artificial patch, which extends across the pulmonary valve.
- Infants who are very cyanosed in the neonatal period require a shunt to increase pulmonary blood flow. This is usually done by surgical placement of an artificial tube between the subclavian artery and the pulmonary artery (a modified Blalock–Taussig shunt), or sometimes by balloon dilatation of the right ventricular outflow tract.
- Hypercyanotic spells are usually self-limiting and followed by a period of sleep. If prolonged (beyond about 15 min), they require prompt treatment with:
  - sedation and pain relief (morphine is excellent)
  - intravenous propranolol (or an α adrenoceptor agonist), which probably works both as a peripheral vasoconstrictor and by relieving the subpulmonary muscular obstruction that is the cause of reduced pulmonary blood flow
  - intravenous volume administration
  - bicarbonate to correct acidosis
  - muscle paralysis and artificial ventilation in order to reduce metabolic oxygen demand.

Transposition of the great arteries

The aorta is connected to the right ventricle, and the pulmonary artery is connected to the left ventricle (discordant ventriculo–arterial connection). The blue blood is therefore returned to the body and the pink blood is returned to the lungs (Fig. 17.9a). There are two parallel circulations – unless there is mixing of blood between them this condition is incompatible with life. Fortunately, there are a number of naturally occurring associated anomalies, e.g. VSD, ASD and PDA, as well as therapeutic interventions which can achieve this mixing in the short term.

Clinical features

Symptoms

Cyanosis is the predominant symptom. It may be profound and life-threatening. Presentation is usually on day 2 of life when ductal closure leads to a marked reduction in mixing of the desaturated and saturated blood. Cyanosis will be less severe and presentation delayed if there is more mixing of blood from associated anomalies, e.g. an ASD.

Physical signs (Fig. 17.9b)

- Cyanosis is always present
- The second heart sound is often loud and single
- Usually no murmur, but may be a systolic murmur from increased flow or stenosis within the left ventricular (pulmonary) outflow tract.

Investigations

Chest radiograph (Fig. 17.9c)

This may reveal the classic findings of a narrow upper mediastinum with an 'egg on side' appearance of the cardiac shadow (due to the anteroposterior relationship of the great vessels, narrow vascular pedicle and hypertrophied right ventricle, respectively). Increased pulmonary vascular markings are common due to *increased* pulmonary blood flow.

ECG (Fig. 17.9d)

This is usually normal.

Echocardiography

This is essential to demonstrate the abnormal arterial connections and associated abnormalities.

Management

- In the sick cyanosed neonate, the key is to improve mixing.
- Maintaining the patency of the ductus arteriosus with a prostaglandin infusion is mandatory.
- A balloon atrial septostomy may be a life-saving procedure which may need to be performed in 20% of those with transposition of the great arteries (Fig. 17.9e–g). A catheter, with an inflatable balloon at its tip, is passed through the umbilical or femoral vein and then on through the right atrium and foramen ovale. The balloon is inflated within the left atrium and then pulled back through the atrial septum. This tears the atrial septum, renders the flap valve of the foramen ovale incompetent, and so allows mixing of the systemic and pulmonary venous blood within the atrium.
- All patients with transposition of the great arteries will require surgery, which is usually the arterial switch procedure in the neonatal period. In this operation, performed in the first few days of life, the pulmonary artery and aorta are transected

Transposition of the great arteries

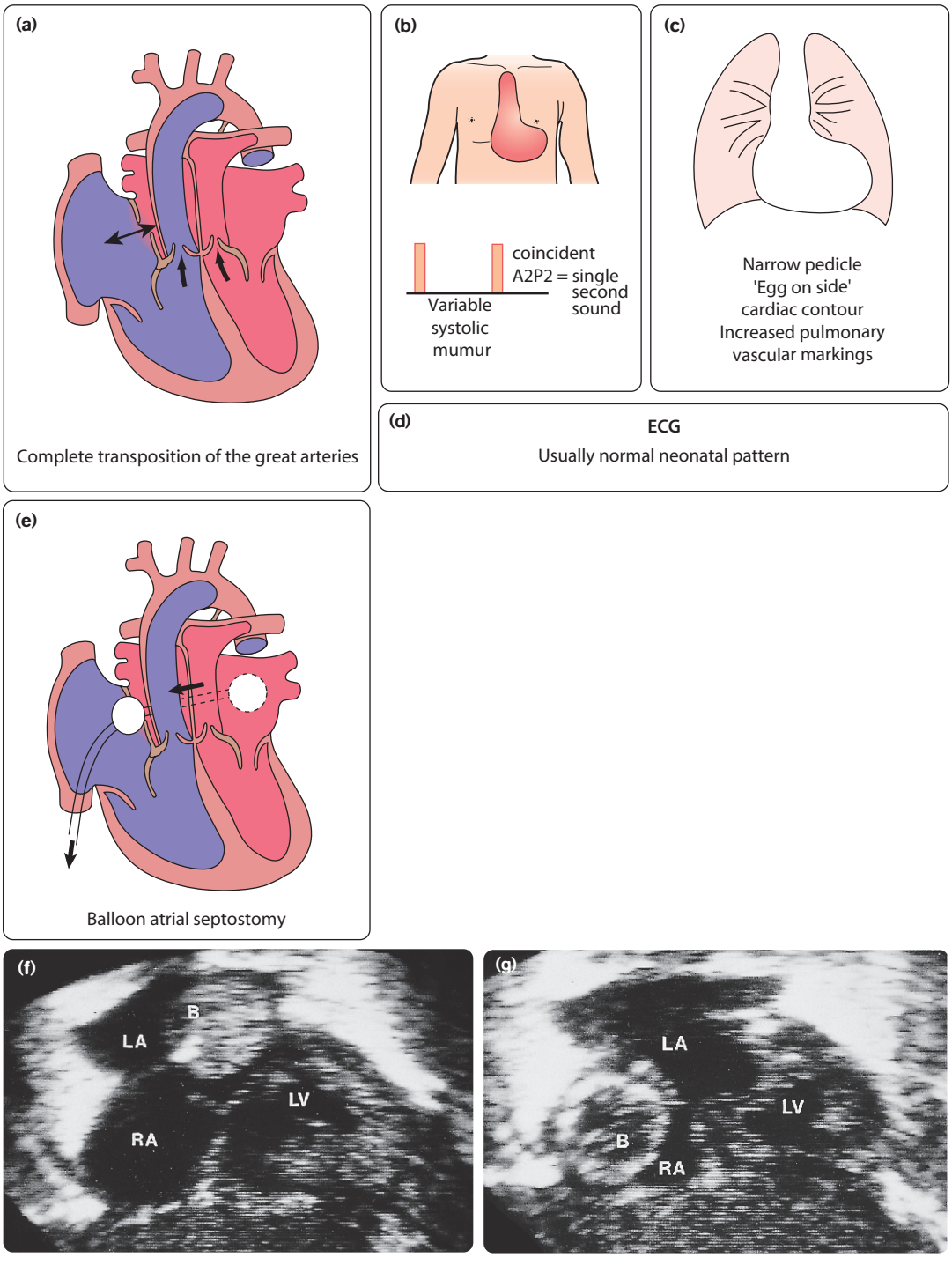


Figure 17.9 Transposition of the great arteries. **(a)** Transposition of the great arteries. There must be mixing of blood between the two circulations for this to be compatible with life. **(b)** Heart sounds. **(c)** Chest radiograph. **(d)** ECG. **(e)** Balloon atrial septostomy. A balloon (about 2 ml) is pulled through the atrial septum from the left atrium to the right atrium in order to increase the size of the atrial defect. This is done with echocardiographic guidance. **(f)** Echocardiogram showing balloon in left atrium. **(g)** Balloon has been pulled through the atrial septum and is now in the right atrium. (B, balloon; LA, left atrium; RA, right atrium; LV, left ventricle)

Summary

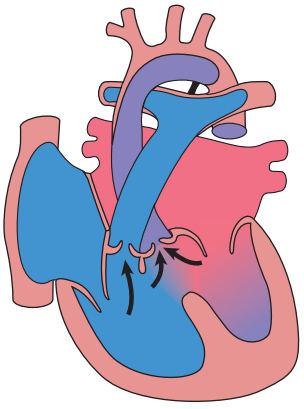
Cyanotic congenital heart disease

Lesion	Clinical features	Management
Tetralogy of Fallot	Loud murmur at ULSE Clubbing of fingers and toes (older) Hypercyanotic spells	Surgery at 6–9 months
Transposition of the great arteries	Neonatal cyanosis No murmur	Prostaglandin infusion Balloon atrial septostomy Arterial switch operation in neonatal period
Eisenmenger syndrome	No murmur Right heart failure (late)	Medication to delay transplantation

above the arterial valves and switched over. In addition, the coronary arteries have to be transferred across to the new aorta.

Eisenmenger syndrome

If high pulmonary blood flow due to a large left-to-right shunt or common mixing is not treated at an early stage, then the pulmonary arteries become thick walled and the resistance to flow increases (Fig. 17.10). Gradually, those children that survive, become less symptomatic, as the shunt decreases. Eventually, at about 10–15 years, the shunt reverses and the teenager becomes blue, which is Eisenmenger syndrome. This situation is progressive and the adult will die in right heart failure at a variable age, usually in the fourth or fifth decade of life. Treatment is aimed at prevention of this condition, with early intervention for high pulmonary blood flow. Transplantation is not easily available although there are now medicines to palliate such pulmonary vascular disease (see Pulmonary hypertension, below).



Eisenmenger syndrome

Figure 17.10 Eisenmenger syndrome with right-to-left shunting from pulmonary vascular disease following increased pulmonary blood flow and pulmonary hypertension with large VSD.

Common mixing (blue and breathless)

These include:

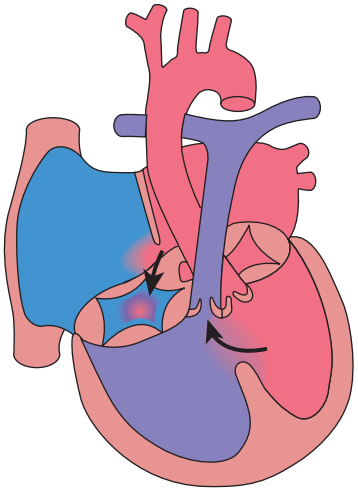
- Atrioventricular septal defect (complete)
- Complex congenital heart disease e.g. tricuspid atresia

Atrioventricular septal defect (complete)

This occurs in:

- Atrioventricular septal defect (complete)
- Complex congenital heart disease

This is most commonly seen in children with Down syndrome (Fig. 17.11). A complete atrioventricular septal defect (cAVSD) is a defect in the middle of the heart with a single five-leaflet (common) valve between



Atrioventricular septal defect

Figure 17.11 Complete atrioventricular septal defect, with a common atrioventricula valve between a large atrial and ventricular component to the AVSD.

Summary

Common mixing

Lesion	Clinical features	Management
Atrioventricular septal defect (complete)	Down syndrome (often) Cyanosis at birth Breathless at 2–3 weeks of life	Treat heart failure medically Surgical repair at 3 months
Complex diseases (e.g. tricuspid atresia)	Cyanosis Breathless	Shunt (Blalock–Taussig) or pulmonary artery banding, then surgery (Fontan operation)

the atria and ventricles which stretches across the entire atrioventricular junction and tends to leak. As there is a large defect there is pulmonary hypertension.

Features of a complete atrioventricular septal defect are:

- Presentation on antenatal ultrasound screening
- Cyanosis at birth or heart failure at 2–3 weeks of life
- No murmur heard, the lesion being detected on routine echocardiography screening in a newborn baby with Down syndrome
- There is always a superior axis on the ECG
- Management is to treat heart failure medically (as for large VSD) and surgical repair at 3–6 months of age.

Complex congenital heart disease

It is difficult to generalise about these conditions, (tricuspid atresia, mitral atresia, double inlet left ventricle, common arterial trunk – truncus arteriosus) since their main presenting feature depends on whether cyanosis or heart failure is more predominant. Tricuspid atresia is the commonest.

Tricuspid atresia

In tricuspid atresia (Fig. 17.12), only the left ventricle is effective, the right being small and non-functional.

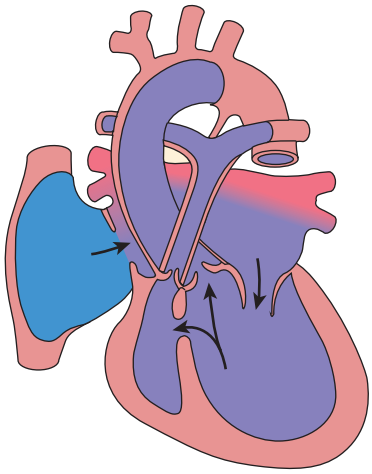
Clinical features

There is 'common mixing' of systemic and pulmonary venous return in the left atrium. Presentation is with cyanosis in the newborn period if duct dependent, or the child may be well at birth and become cyanosed or breathless.

Management

Early palliation (as with all the common mixing complex diseases) is performed to maintain a secure supply of blood to the lungs at low pressure, by:

- A Blalock–Taussig shunt insertion (between the subclavian and pulmonary artery) in children who are severely cyanosed
- Pulmonary artery banding operation to reduce pulmonary blood flow if breathless.



Tricuspid atresia

Figure 17.12 In tricuspid atresia, there is only one effective ventricle because of complete absence of the tricuspid valve.

Completely corrective surgery is not possible with most, as there is often only one effective functioning ventricle. Palliation is performed (Glenn or hemi-Fontan operation connecting the superior vena cava to the pulmonary artery after 6 months of age and a Fontan operation to also connect the inferior vena cava to the pulmonary artery at 3–5 years).

Thus, the left ventricle drives blood around the body and systemic venous pressure supplies blood to the lungs. The Fontan operation results in a less than ideal functional outcome, but has the advantages of relieving cyanosis and removing the long-term volume load on the single functional ventricle.

Outflow obstruction in the well child

These lesions are:

- Aortic stenosis
- Pulmonary stenosis
- Adult-type coarctation of the aorta.

Aortic stenosis

The aortic valve leaflets are partly fused together, giving a restrictive exit from the left ventricle (Fig. 17.13a). There may be one to three aortic leaflets. Aortic stenosis may not be an isolated lesion. It is often associated with mitral valve stenosis and coarctation of the aorta, and their presence should always be excluded.

Clinical features

Most present with an asymptomatic murmur. Those with severe stenosis may present with reduced exercise tolerance, chest pain on exertion or syncope.

In the neonatal period, those with *critical* aortic stenosis and a duct-dependent systemic circulation may present with severe heart failure leading to shock.

Physical signs (Fig. 17.13b)

- Small volume, slow rising pulses
- Carotid thrill (always)
- Ejection systolic murmur maximal at the upper right sternal edge radiating to the neck
- Delayed and soft aortic second sound
- Apical ejection click.

Investigations

Chest radiograph (Fig. 17.13c)

Normal or prominent left ventricle with post-stenotic dilatation of the ascending aorta.

ECG (Fig. 17.13d)

There may be left ventricular hypertrophy.

Management

In children, regular clinical and echocardiographic assessment is required in order to assess when to intervene. Children with symptoms on exercise or who have a high resting pressure gradient (>64 mmHg) across the aortic valve will undergo balloon valvotomy. Balloon dilatation in older children is generally safe and uncomplicated, but in neonates this is much more difficult and dangerous.

Most neonates and children with significant aortic valve stenosis requiring treatment in the first few years of life will eventually require aortic valve replacement. Early treatment is therefore palliative and directed towards delaying this for as long as possible.

Pulmonary stenosis

The pulmonary valve leaflets are partly fused together, giving a restrictive exit from the right ventricle.

Clinical features

Most are asymptomatic (Fig. 17.14a). It is diagnosed clinically. A small number of neonates with *critical* pulmonary stenosis have a duct-dependent pulmonary circulation and present in the first few days of life with cyanosis.

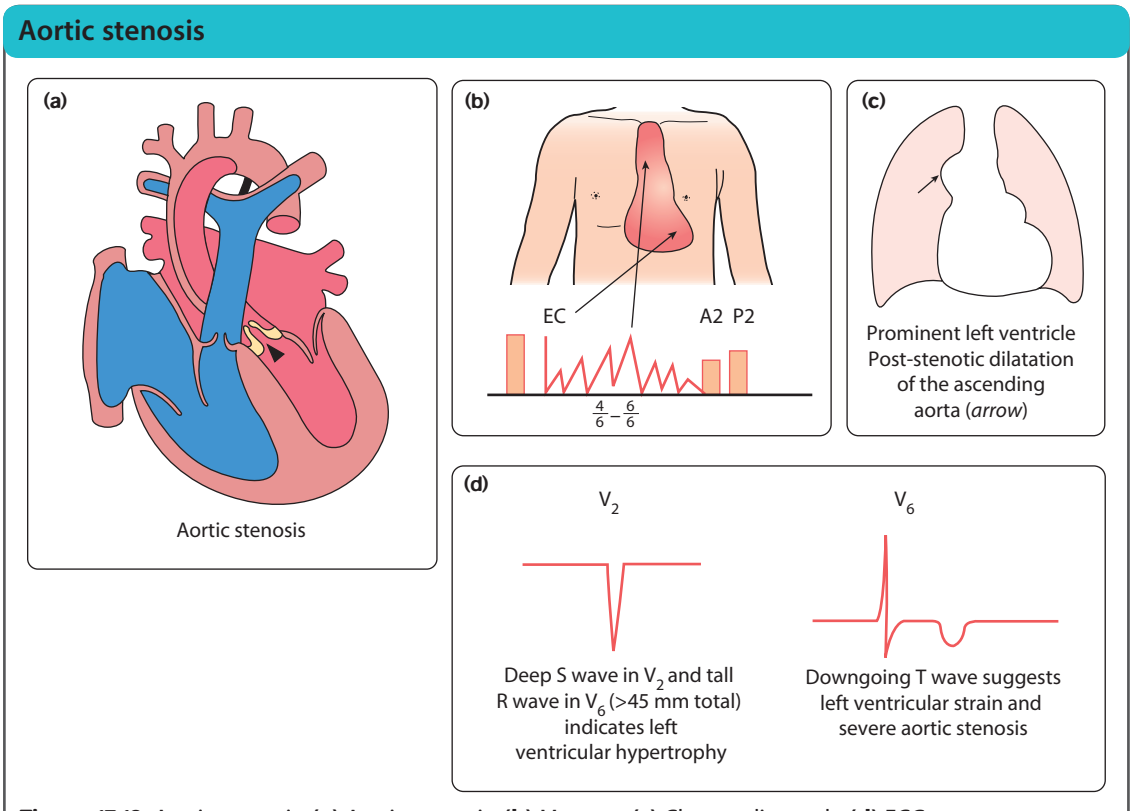


Figure 17.13 Aortic stenosis. **(a)** Aortic stenosis. **(b)** Murmur. **(c)** Chest radiograph. **(d)** ECG.

Physical signs (Fig. 17.14b)

- An ejection systolic murmur best heard at the upper left sternal edge; thrill may be present
- An ejection click best heard at the upper left sternal edge
- When severe, there is a prominent right ventricular impulse (heave).

Investigations

Chest radiograph (Fig. 17.14c)

Normal or post-stenotic dilatation of the pulmonary artery.

ECG (Fig. 17.14d)

Shows evidence of right ventricular hypertrophy (upright T wave in V_1).

Management

Most children are asymptomatic and when the pressure gradient across the pulmonary valve becomes markedly increased (> about 64 mmHg), intervention will be required. Trans-catheter balloon dilatation is the treatment of choice in most children.

Adult-type coarctation of the aorta

This uncommon lesion (Fig. 17.15a) is not duct dependent. It gradually becomes more severe over many years.

Clinical features (Fig 17.15b)

- Asymptomatic
- Systemic hypertension in the right arm
- Ejection systolic murmur at upper sternal edge
- Collaterals heard with continuous murmur at the back
- Radio-femoral delay. This is due to blood bypassing the obstruction via collateral vessels in the chest wall and hence the pulse in the legs is delayed.

Investigations

Chest radiograph (Fig. 17.15c)

- 'Rib notching' due to the development of large collateral intercostal arteries running under the ribs posteriorly to bypass the obstruction
- '3' sign, with visible notch in the descending aorta at site of the coarctation

ECG

- Left ventricular hypertrophy (Fig. 17.15d).

Management

When the condition becomes severe, as assessed by echocardiography, a stent may be inserted at cardiac catheter. Sometimes surgical repair is required.

Pulmonary stenosis

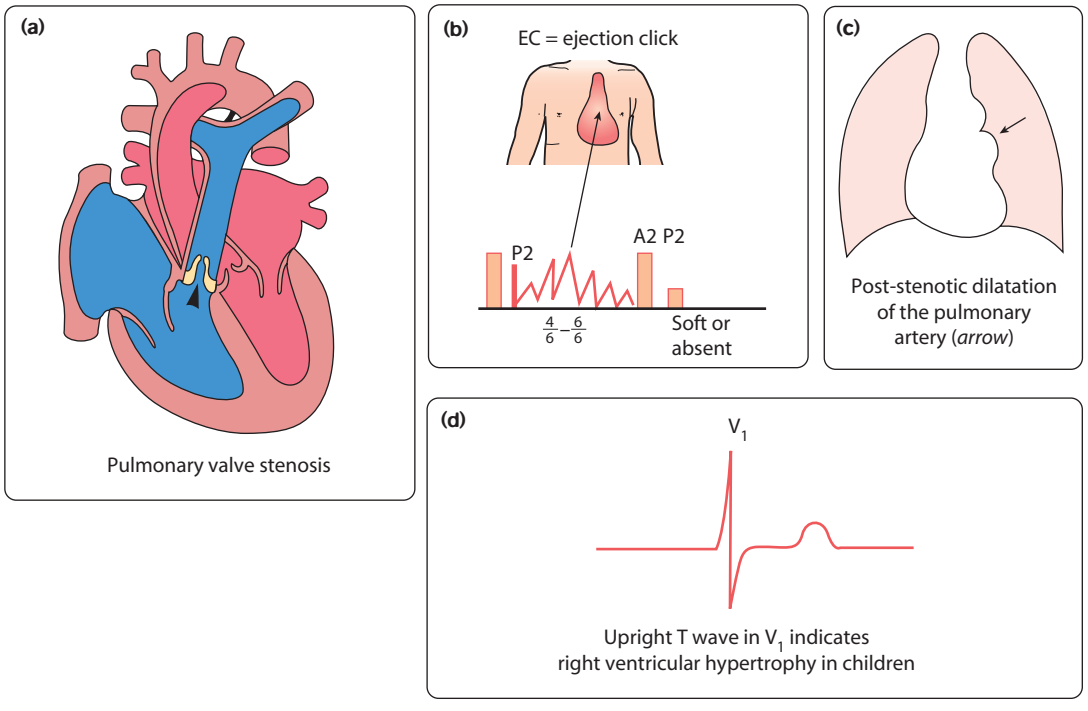


Figure 17.14 Pulmonary valve stenosis. **(a)** Pulmonary valve stenosis. **(b)** Murmur. **(c)** Chest radiograph. **(d)** ECG.

Summary

Outflow obstruction in the well child

Lesion	Signs	Management
Aortic stenosis	Murmur, upper right sternal edge; carotid thrill	Balloon dilatation
Pulmonary stenosis	Murmur, upper left sternal edge; no carotid thrill	Balloon dilatation
Coarctation (adult type)	Systemic hypertension Radio-femoral delay	Stent insertion or surgery

Coarctation of the aorta

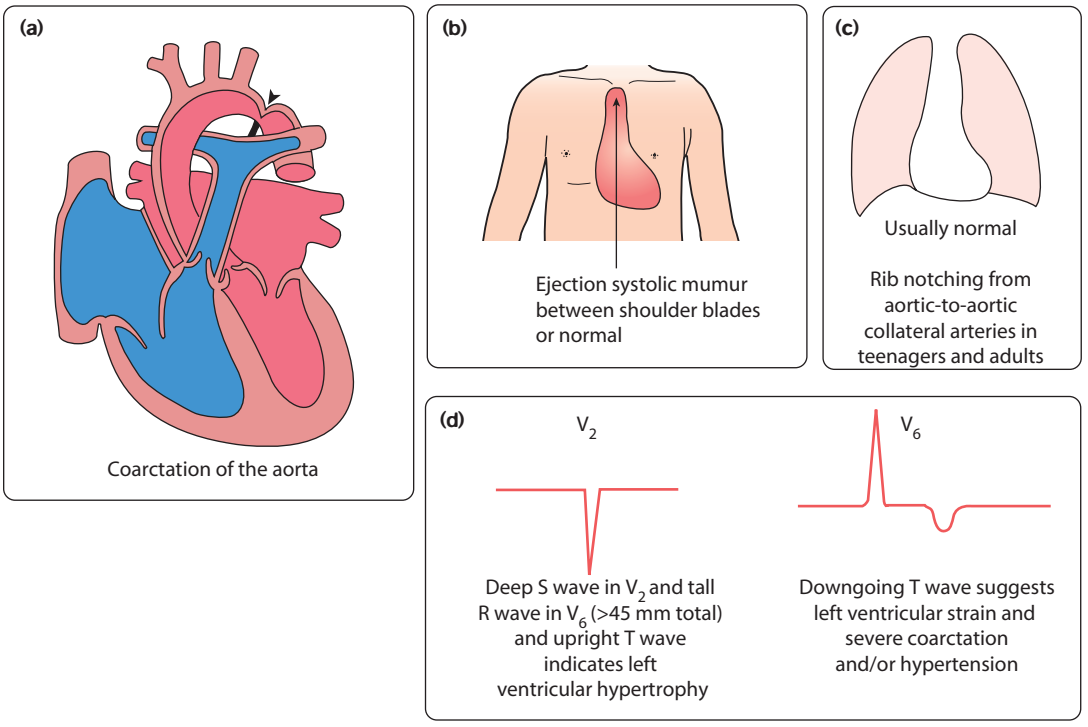


Figure 17.15 Coarctation of the aorta. **(a)** Coarctation of the aorta. There is narrowing of the aorta distal to the left subclavian artery adjacent to the insertion of the arterial duct. **(b)** Murmur. **(c)** Chest radiograph. **(d)** ECG.

Outflow obstruction in the sick infant

These lesions include:

- Coarctation of the aorta
- Interruption of the aortic arch
- Hypoplastic left heart syndrome.

Clinical features are:

- In all of these children, they usually present sick with heart failure and shock in the neonatal period, unless diagnosed on antenatal ultrasound
- Management is to resuscitate first (ABC)
- Prostaglandin should be commenced at the earliest opportunity
- Referral is made to a cardiac centre for early surgical intervention.

Coarctation of the aorta (Fig. 17.3)

This is due to arterial duct tissue encircling the aorta just at the point of insertion of the duct. When the duct closes, the aorta also constricts, causing severe obstruction to the left ventricular outflow. This is the commonest cause of collapse due to left outflow obstruction.

Clinical features

Examination on the first day of life is usually normal. The neonates usually present with acute circulatory collapse at 2 days of age when the duct closes.

Physical signs

- A sick baby, with severe heart failure
- Absent femoral pulses
- Severe metabolic acidosis.

Investigations

Chest radiograph

Cardiomegaly from heart failure and shock.

ECG

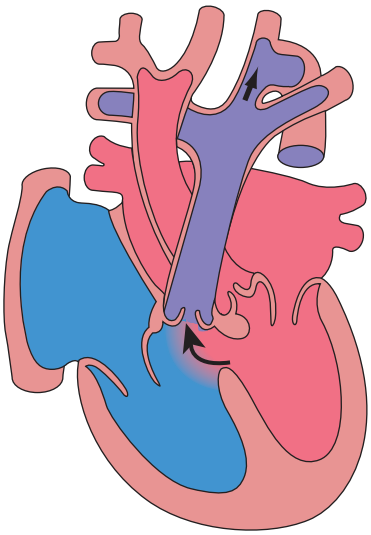
Normal.

Management

As for all the children in this section with an obstructed left outflow tract (see above). Surgical repair is performed soon after diagnosis.

Interruption of the aortic arch

- Uncommon, with *no* connection between the proximal aorta and distal to the arterial duct, so that the cardiac output is dependent on right-to-left shunt via the duct Fig. 17.16



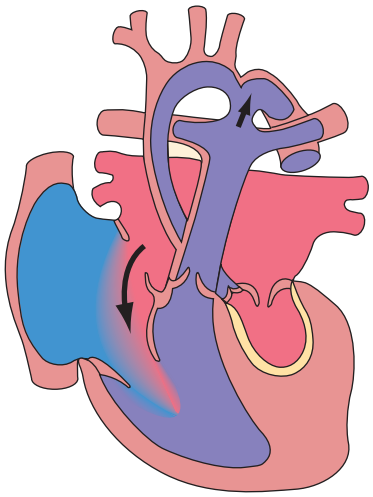
Interrupted aortic arch

Figure 17.16 Interruption of the aortic arch. The lower body circulation is maintained by right-to-left flow of blood across the duct.

- A VSD is usually present
- Presentation is with shock in the neonatal period as above
- Complete correction with closure of the VSD and repair of the aortic arch is usually performed within the first few days of life.
- Association with other conditions (DiGeorge syndrome – absence of thymus, palatal defects, immunodeficiency and hypocalcaemia and 22q11.2 gene micro-deletion).

Hypoplastic left heart syndrome

In this condition there is underdevelopment of the entire left side of the heart (Fig. 17.17). The mitral valve is small or atretic, the left ventricle is diminutive and there is usually aortic valve atresia. The ascending aorta is very small, and there is almost invariably coarctation of the aorta.



Hypoplastic left heart

Figure 17.17 Hypoplastic left heart syndrome. The entire left side of the heart is underdeveloped.

Summary

Left heart outflow obstruction in the sick infant – duct-dependent lesions

Lesion	Clinical features	Management
Coarctation of the aorta	Circulatory collapse	Maintain ABC
	Absent femoral pulses	Prostaglandin infusion
Interruption of the aortic arch	Circulatory collapse	Maintain ABC
	Absent femoral pulses and absent left brachial pulse	Prostaglandin infusion
Hypoplastic left heart syndrome	Circulatory collapse	Maintain ABC
	All peripheral pulses absent	Prostaglandin infusion

Clinical features

These children may be detected antenatally at ultrasound screening. This allows for effective counselling and prevents the child from becoming sick after birth. If they do present after birth, they are the sickest of all neonates presenting with a duct-dependent systemic circulation. There is no flow through the left side of the heart, so ductal constriction leads to profound acidosis and rapid cardiovascular collapse. There is weakness or absence of all peripheral pulses, in contrast to weak femoral pulses in coarctation of the aorta.

Management

The management of this condition consists of a difficult neonatal operation called the Norwood procedure. This is followed by a further operation (Glenn or hemi-Fontan) at about 6 months and again (Fontan) at about 3 years.

Care following cardiac surgery

Most children recover rapidly following cardiac surgery and are back at nursery or school within a month. Exercise tolerance will be variable and most children can be allowed to find their own limits. Restricted exercise is advised only for children with severe residual aortic stenosis and for ventricular dysfunction.

Most of the children are followed up in specialist cardiac clinics. Most lead normal, unrestricted lives, but any change in symptoms, e.g. decreasing exercise tolerance or palpitations, requires further investigation. An increasing number of adolescents and young adults require revision of surgery performed in early life. The most common reason for this is replacement of artificial valves and relief of post-surgical suture line stenosis, for example re-coarctation or pulmonary artery stenosis.

Cardiac arrhythmias

Sinus arrhythmia is normal in children and is detectable as a cyclical change in heart rate with respiration. There is acceleration during inspiration and slowing on expiration (the heart rate changing by up to 30 beats/min).

Supraventricular tachycardia

This is the most common childhood arrhythmia. The heart rate is rapid, between 250 and 300 beats/min. It can cause poor cardiac output and pulmonary oedema. It typically presents with symptoms of heart failure in the neonate or young infant. It is a cause of *hydrops fetalis* and intrauterine death. The term re-entry tachycardia is used because a circuit of conduction is set up, with premature activation of the atrium via an accessory pathway. There is rarely a structural heart problem, but an echocardiogram should be performed.

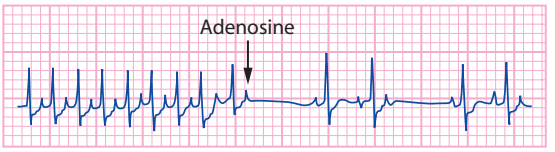


Figure 17.18 Rhythm strip showing supraventricular re-entry tachycardia, in which there is a narrow complex tachycardia (<120 ms or three small squares) of 250–300 beats/min, and response to treatment with adenosine.

Investigation

The ECG will generally show a narrow complex tachycardia of 250–300 beats/min (Fig. 17.18). It may be possible to discern a P wave after the QRS complex due to retrograde activation of the atrium via the accessory pathway. If heart failure is severe, there may be changes suggestive of myocardial ischaemia, with T-wave inversion in the lateral precordial leads. When in sinus rhythm, a short P–R interval may be discernible. In the Wolff–Parkinson–White (WPW) syndrome, the early antegrade activation of the ventricle via the pathway results in a short P–R interval and a delta wave.

Management

In the severely ill child, prompt restoration of sinus rhythm is the key to improvement. This is achieved by:

- Circulatory and respiratory support – tissue acidosis is corrected, positive pressure ventilation if required
- Vagal stimulating manoeuvres, e.g. carotid sinus massage or cold ice pack to face, successful in about 80%
- Intravenous adenosine – the treatment of choice. This is safe and effective, inducing atrioventricular block after rapid bolus injection. It terminates the tachycardia by breaking the re-entry circuit that is set up between the atrioventricular node and accessory pathway. It is given incrementally in increasing doses
- Electrical cardioversion with a synchronised DC shock (0.5–2 J/kg body weight) if adenosine fails.

Once sinus rhythm is restored, maintenance therapy will be required, e.g. with flecainide or sotalol. Digoxin can be used on its own when there is no overt pre-excitation wave (delta wave) on the resting ECG, but propranolol can be added in the presence of pre-excitation. Even though the resting ECG may remain abnormal, 90% of children will have no further attacks after infancy. Treatment is therefore stopped at 1 year of age. Those who have WPW syndrome need to be assessed to ensure they cannot conduct quickly and this may be undertaken in teenage life, with atrial pacing. This will reduce the small chance of sudden death in such patients. Those who relapse or are at risk are usually treated with percutaneous radiofrequency ablation or cryoablation of the accessory pathway.

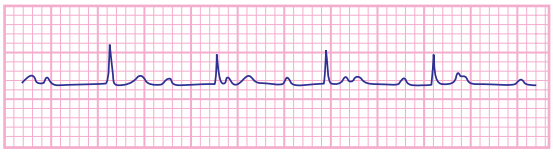


Figure 17.19 ECG of congenital complete heart block. The P waves and QRS complexes are dissociated.

Congenital complete heart block

This is a rare condition (Fig. 17.19) which is usually related to the presence of anti-Ro or anti-La antibodies in maternal serum. These mothers will have either manifest or latent connective tissue disorders. Subsequent pregnancies are often affected. This antibody appears to prevent normal development of the electrical conduction system in the developing heart, with atrophy and fibrosis of the atrioventricular node. It may cause fetal hydrops, death in utero and heart failure in the neonatal period. However, most remain symptom-free for many years, but a few become symptomatic with presyncope or syncope. All children with symptoms require insertion of an endocardial pacemaker. There are other rare causes of complete heart block.

Other arrhythmias

Long QT syndrome may be associated with sudden loss of consciousness during exercise, stress or emotion, usually in late childhood. It may be mistakenly diagnosed as epilepsy. If unrecognised, sudden death from ventricular tachycardia may occur. Inheritance is autosomal dominant; there are several phenotypes. It has been associated with erythromycin therapy, electrolyte disorders and head injury.

It is one of the group of channelopathies caused by specific gene mutations. Abnormalities of the sodium, potassium or calcium channels lead to gain or loss of function. Anyone with a family history of sudden unexplained death, or a history of syncope on exertion should be assessed.

Atrial fibrillation, atrial flutter, ectopic atrial tachycardia, ventricular tachycardia and ventricular fibrillation occur in children, but all are rare. They are most often seen in children who have undergone surgery for complex congenital heart disease.

Syncope

This is a common symptom in teenagers and usually does not represent cardiac disease.

The causes are:

- Neurocardiogenic – prolonged standing and vagal symptoms
- Situational – defecation, urination, cough or swallowing
- Orthostatic – BP fall > 20 mmHg after 3 min
- Ischaemic
- Arrhythmic – heart block, supraventricular tachycardia, ventricular tachycardia.

In most, the cause is non-cardiac. Features suggestive of a cardiac cause are:

- Symptoms on exercise – potentially dangerous
- Family history of sudden unexplained death
- Palpitations.

Check blood pressure and signs of cardiac disease (murmur, femoral pulses, Marfan syndrome).

Investigate with an ECG.

Chest pains

Rarely due to cardiac disease in children. Only those occurring with palpitations or on exertion suggest a possible cardiac origin.

Rheumatic fever

This is now rare in the developed world, but remains the most important cause of heart disease in children worldwide. Improvements in sanitation, social factors, the more liberal use of antibiotics and changes in streptococcal virulence have led to its virtual disappearance in developed countries. In susceptible individuals, there is an abnormal immune response to a preceding infection with group A β-haemolytic streptococcus. The disease mainly affects children aged 5–15 years.

Clinical features

After a latent interval of 2–6 weeks following a pharyngeal infection, polyarthritis, mild fever and malaise develop. The clinical features and diagnostic criteria are shown in Figure 17.20.

Chronic rheumatic heart disease

The most common form of long-term damage from scarring and fibrosis of the valve tissue of the heart is mitral stenosis. If there have been repeated attacks of rheumatic fever with carditis, this may occur as early as the second decade of life, but usually symptoms do not develop until early adult life. Although the mitral valve is the most frequently affected, aortic, tricuspid and, rarely, pulmonary valve disease may occur.

Management

The acute episode is usually treated with bed rest and anti-inflammatory agents. While there is evidence of active myocarditis (echocardiographic changes with a raised ESR), bed rest and limitation of exercise are essential. Aspirin is very effective at suppressing the inflammatory response of the joints and heart. It needs to be given in high dosage and serum levels monitored. If the fever and inflammation do not resolve rapidly, corticosteroids will be required. Symptomatic heart failure is treated with diuretics and ACE inhibitors, and significant pericardial effusions will require pericardiocentesis. Anti-streptococcal antibiotics may be given if there is any evidence of persisting infection.

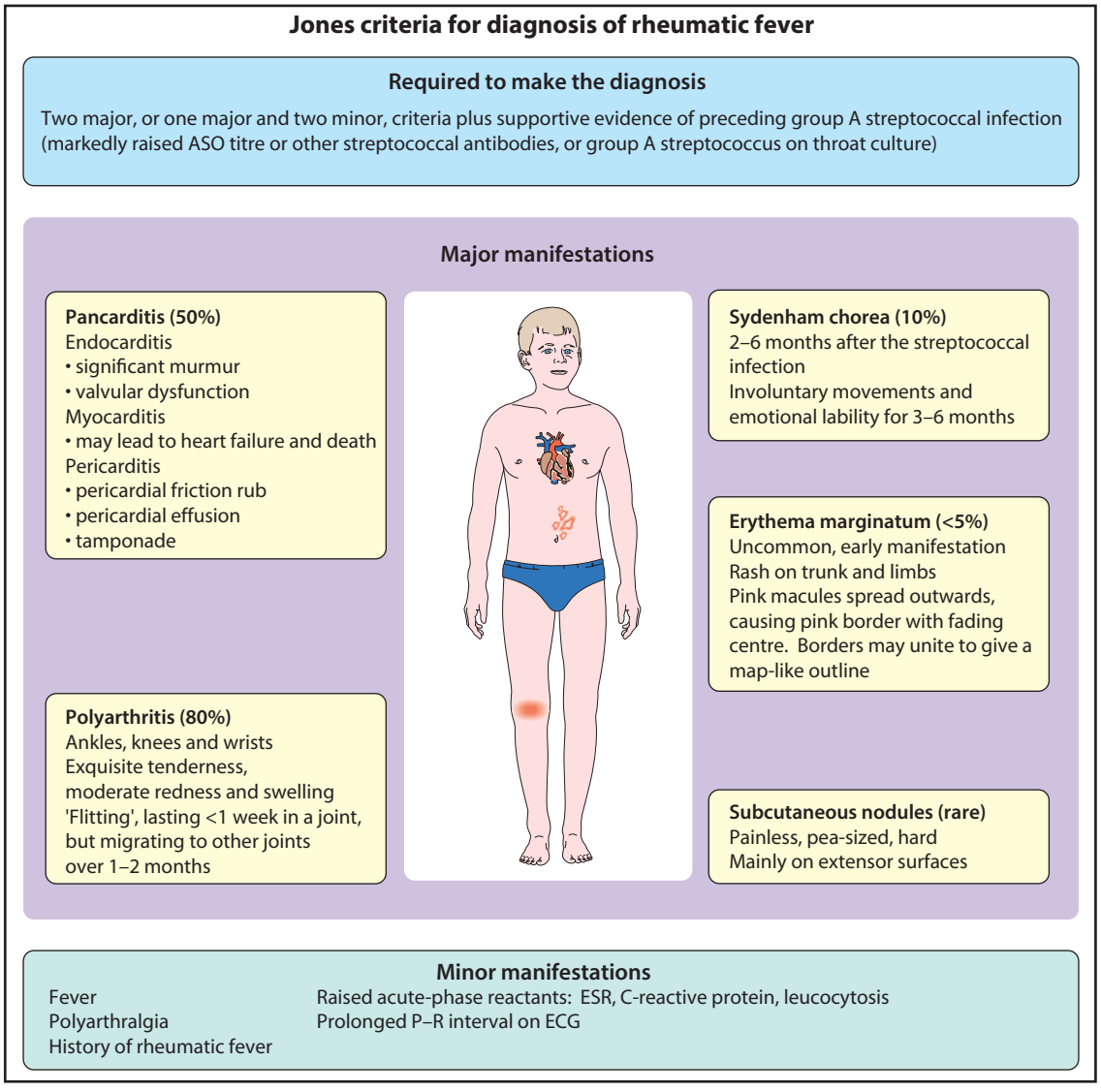


Figure 17.20 Jones criteria for diagnosis of rheumatic fever.

Following resolution of the acute episode, recurrence should be prevented. Monthly injections of benzathine penicillin is the most effective prophylaxis. Alternatively, the penicillin can be given orally every day, but compliance may be a problem. Oral erythromycin can be substituted in those sensitive to penicillin. The length of treatment is controversial. Most recommend treatment to the age of 18 or 21 years, but, more recently, lifelong prophylaxis has been advocated. The severity of eventual rheumatic valvular disease relates to the number of childhood episodes of rheumatic fever.

Infective endocarditis

All children of any age with congenital heart disease (except secundum ASD), including neonates, are at risk of infective endocarditis. The risk is highest when there is a turbulent jet of blood, as with a VSD, coarctation of the aorta and persistent ductus arteriosus or if prosthetic material has been inserted at surgery. It may be difficult to diagnose, but should be suspected in any child or adult with a sustained fever, malaise, raised ESR, unexplained anaemia or haematuria. The presence of the classical peripheral stigmata of infective endocarditis should not be relied upon.

Clinical signs

- Fever
- Anaemia and pallor
- Splinter haemorrhages in nailbed
- Clubbing (late)
- Necrotic skin lesions (Fig. 17.21)
- Changing cardiac signs
- Splenomegaly
- Neurological signs from cerebral infarction
- Retinal infarcts
- Arthritis/arthralgia
- Haematuria (microscopic).



Figure 17.21 Widespread infected emboli and infarcts in a child with bacterial endocarditis. The tip of the third toe is gangrenous.

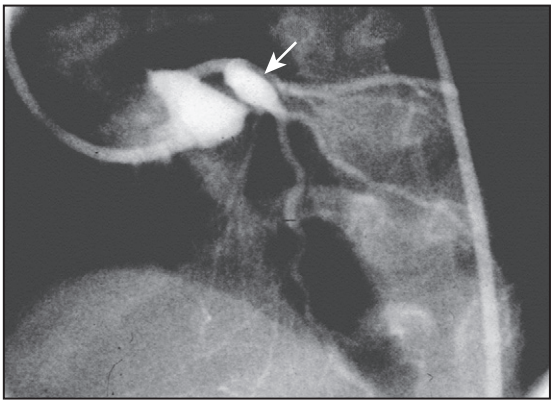


Figure 17.22 Kawasaki disease. Angiogram showing coronary artery aneurysm.

Diagnosis

Multiple blood cultures should be taken before antibiotics are started. Detailed cross-sectional echocardiography may confirm the diagnosis by identification of vegetations but can never exclude it. The vegetations consist of fibrin and platelets and contain infecting organisms. Acute-phase reactants are raised and can be useful to monitor response to treatment.

The most common causative organism is α-haemolytic streptococcus (*Streptococcus viridans*). Bacterial endocarditis is usually treated with high-dose penicillin in combination with an aminoglycoside, giving 6 weeks of intravenous therapy and checking that the serum level of the antibiotic will kill the organism. If there is infected prosthetic material, e.g. prosthetic valves, VSD patches or shunts, there is less chance of complete eradication and surgical removal may be required.

Prophylaxis

The most important factor in prophylaxis against endocarditis is good dental hygiene, and this should be strongly encouraged in all children with congenital heart disease.

Antibiotic prophylaxis is *no longer recommended in the UK*, but may be required in other countries for:

- Dental treatment, however trivial
- Surgery, which is likely to be associated with bacteraemia.

Myocarditis/cardiomyopathy

Dilated cardiomyopathy (a large, poorly contracting heart) may be inherited, secondary to metabolic disease or may result from a direct viral infection of the myocardium. It should be suspected in any child with an enlarged heart and heart failure who has previously been well. The diagnosis is readily made on echocardiography. Treatment is symptomatic with diuretics and ACE inhibitors and carvedilol, a β-adrenoceptor blocking agent. The role of steroids and immunoglobulin infusion is controversial. Myocarditis usually improves spontaneously, but some children ultimately require heart transplantation. Other cardiomyopathies (hypertrophic/restrictive) are rare in childhood and are usually related to a systemic disease (e.g. Hurler, Pompe or Noonan syndromes).

Kawasaki disease

This mainly affects children of 6 months to 5 years. Clinical features are described in Chapter 14. It is uncommon but can cause significant cardiac disease. An echocardiogram is performed at diagnosis which may show a pericardial effusion, myocardial disease (poor contractility), endocardial disease (valve regurgitation) or coronary disease with aneurysm formation, which can be giant (>8 mm in diameter). If the coronary arteries are abnormal, angiography (Fig. 17.22) or MRI will be required.

Pulmonary hypertension

This is of increasing importance in paediatric cardiology, as there is now effective medication for most causes. It can be caused by a number of different diseases (Box 17.4). From the cardiac perspective, most children with pulmonary hypertension (high pulmonary artery pressure, mean >25 mmHg), have a large post-tricuspid shunt with high pulmonary blood flow and low resistance, e.g. VSD, AVSD or PDA. The pressure falls to normal if the defect is corrected by surgery within 6 months of age. If these children are left untreated, however, the high flow and pressure causes irreversible damage to the pulmonary vascular bed (pulmonary vascular disease), which is not correctable other than by heart/lung transplantation.

Many medical therapies are now available, which may act on the pulmonary vasculature on the cyclic

Box 17.4 Causes of pulmonary hypertension

- Pulmonary arterial hypertension
  - Idiopathic: sporadic or familial
  - Post-tricuspid shunts (e.g. VSD, AVSD, PDA)
  - HIV infection
  - Persistent pulmonary hypertension of the newborn
- Pulmonary venous hypertension
  - Left-sided heart disease
  - Pulmonary vein stenosis or compression
- Pulmonary hypertension with respiratory disease
  - Chronic obstructive or premature lung disease
  - Interstitial lung disease
  - Obstructive sleep apnoea or upper airway obstruction
- Pulmonary thromboembolic disease
- Pulmonary inflammatory or capillary disease.

GMP (guanosine monophosphate) pathway (e.g. inhaled nitric oxide, intravenous magnesium sulphate and oral phosphodiesterase inhibitors including sildenafil) or on the cyclic AMP pathway (intravenous prostacyclin or inhaled iloprost). In addition, endothelin receptor antagonists are valuable but expensive therapy, e.g. oral Bosentan. Anticoagulation is often given with heparin, aspirin or warfarin. These medications allow transplantation to be delayed for many years.

Further reading

Anderson RH, Baker E, Penny D, Redington A, Rigby M, et al: *Paediatric Cardiology*, ed 3. Edinburgh, 2010, Churchill Livingstone.

Website (Accessed May 2011)

Children's Heart Federation: http://www.childrens-heart-fed.org.uk/.

18

Kidney and urinary tract disorders

Congenital abnormalities	325	Renal masses	341
Urinary tract infection	330	Renal calculi	341
Enuresis	335	Renal tubular disorders	341
Proteinuria	336	Acute kidney injury	342
Haematuria	338	Chronic kidney disease	344
Hypertension	340		

The spectrum of renal disease in children differs from that in adults:

- Many structural abnormalities of the kidneys and urinary tract are identified on antenatal ultrasound screening
- Urinary tract infection, vesicoureteric reflux and urinary obstruction have the potential to damage the growing kidney
- Nephrotic syndrome is usually steroid-sensitive and only rarely leads to chronic renal failure
- Chronic renal disorders and the drugs used to treat them may affect growth and development.

Assessment of the kidneys and urinary tract

The glomerular filtration rate (GFR) is low in the newborn infant and is especially low in premature infants; the GFR at 28 weeks' gestation is only 10% of the term infant. In term infants, the corrected GFR (15–20 ml/min per 1.73 m^2) rapidly rises to 1–2 years of age when the adult rate of 80 to 120 ml/min per 1.73 m^2 is achieved (Fig. 18.1). The assessment of renal function in children is listed in Table 18.1 and the radiological investigations of the kidneys and urinary tract in Table 18.2.

Congenital abnormalities

Before antenatal ultrasound scanning became routine, few congenital abnormalities of the kidneys and urinary tract were diagnosed until they caused symptoms in infancy, childhood or, occasionally, adult life. Now the majority are identified in utero and can be managed prospectively. Abnormalities are identified in 1 in 200–400 births. They are potentially important because they may:

- be associated with abnormal renal development or function
- predispose to postnatal infection
- involve urinary obstruction which requires surgical treatment.

The antenatal detection and early treatment of urinary tract anomalies provide an opportunity to minimise or prevent progressive renal damage. A disadvantage is that minor abnormalities are also detected, most commonly mild unilateral pelvic dilatation, which do not require intervention but may lead to over-investigation, unnecessary treatment and unwarranted parental anxiety.

Anomalies detectable on antenatal ultrasound screening

Absence of both kidneys (*renal agenesis*) – as amniotic fluid is mainly derived from fetal urine, there is severe oligohydramnios resulting in Potter syndrome (Fig. 18.2a, 18.2b), which is fatal.

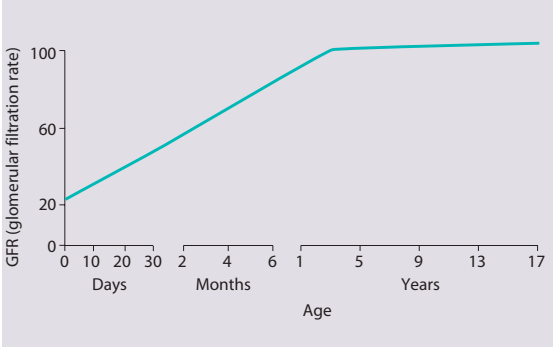


Figure 18.1 Increase in renal function (GFR) with age.

Table 18.1 Assessment of renal function in children

Plasma creatinine concentration (PCr)	Main test of renal function. Rises progressively throughout childhood according to height and muscle bulk. May not be outside laboratory 'normal range' until renal function has fallen to less than half normal
Estimated glomerular filtration rate (eGFR)	The formula eGFR = k × height (cm) ÷ creatinine (μmol/L) provides estimate of GFR. Better measure of renal function than creatinine and useful to monitor renal function serially in children with renal impairment (k is 40 if creatinine measured using Jaffe method or 30 if measured enzymatically)
Inulin or EDTA glomerular filtration rate	More accurate as clearance from the plasma of substances freely filtered at the glomerulus, and is not secreted or reabsorbed by the tubules. Need for repeated blood tests limits use in children
Creatinine clearance	Requires timed urine collection and blood tests. Rarely done in children as inconvenient and inaccurate
Plasma urea concentration	Increased in renal failure, often before creatinine starts rising, and raised levels may be symptomatic. Urea levels also increased by high protein diet and if in a catabolic state.

Table 18.2 Radiological investigation of the kidneys and urinary tract

Ultrasound (US)	Standard imaging procedure of the kidneys and urinary tract, provides anatomical assessment but not function. Excellent at visualising urinary tract dilatation, stones and nephrocalcinosis (small, multiple calcium deposits in renal parenchyma) Advantages: non-invasive, mobile Disadvantages: operator-dependent, will not detect all renal scars
DMSA scan (^{99m}Tc dimercaptosuccinic acid)	*Static* scan of the renal cortex Detects functional defects, such as scars, but very sensitive, so need to wait at least 2 months after a urinary tract infection to avoid diagnosing false 'scars'
Micturating cystourethrogram (MCUG)	Contrast introduced into bladder through urethral catheter. Can visualise bladder and urethral anatomy. Detects vesicoureteric reflux (VUR) and urethral obstruction Disadvantages: invasive and unpleasant investigation especially beyond infancy, high radiation dose
MAG3 renogram (mercapto-acetyl-triglycine, labelled with ^{99m}Tc)	*Dynamic* scan, isotope-labelled substance MAG3 excreted from the blood into the urine. Measures drainage, best performed with a high urine flow. In children old enough to cooperate (usually >4 years), scan during micturition is used to identify VUR
Plain abdominal X-ray	Identifies unsuspected spinal abnormalities May identify renal stones, but poor at showing nephrocalcinosis.

Multicystic dysplastic kidney (*MCDK*) – results from the failure of union of the ureteric bud (which forms the ureter, pelvis, calyces and collecting ducts) with the nephrogenic mesenchyme. It is a non-functioning structure with large fluid-filled cysts with no renal tissue and no connection with the bladder (Fig. 18.3). Half will have involuted by 2 years of age, and nephrectomy is indicated only if it remains very large or hypertension develops, but this is rare. Since they produce no urine, Potter syndrome will result if the lesion is bilateral. Other causes of large cystic kidneys are *autosomal recessive polycystic kidney disease* (*ARPKD*) (Fig. 18.4), *autosomal dominant polycystic kidney disease* (*ADPKD*) (Fig. 18.5) and tuberous sclerosis. In contrast to a multicystic dysplastic kidney, in these disorders some or normal renal function is maintained but both kidneys are always affected. Autosomal dominant polycystic kidney disease has an incidence of 1 in 1000; the main symptoms in childhood are hypertension and haematuria and it causes renal failure in late adulthood.

Some congenital abnormalities of the kidneys and urinary tract

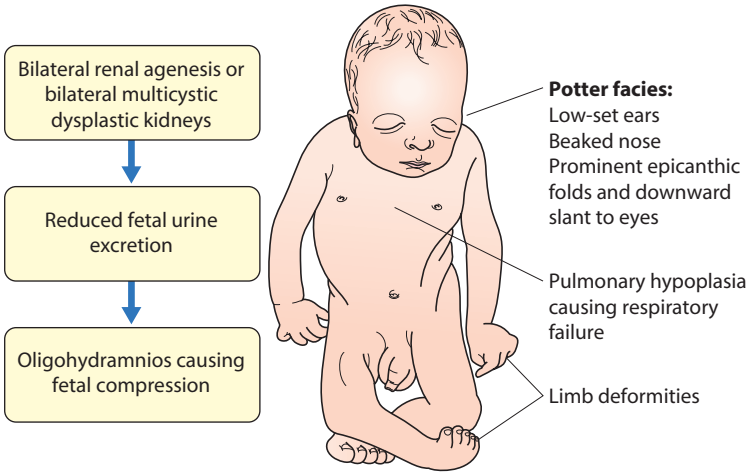


Figure 18.2a Potter syndrome. Intrauterine compression of the fetus from oligohydramnios caused by lack of fetal urine causes a characteristic facies, lung hypoplasia and postural deformities including severe talipes. The infant may be stillborn or die soon after birth from respiratory failure.



Figure 18.2b Facies in Potter syndrome.

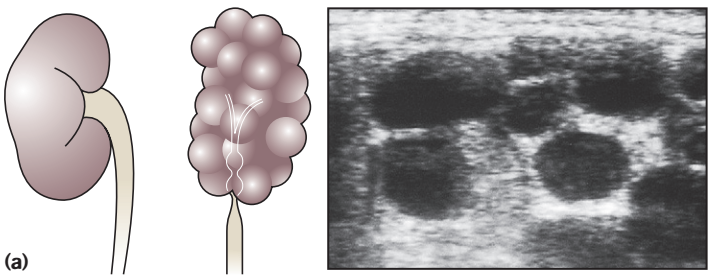


Figure 18.3 **(a)** Multicystic renal dysplasia. The kidney is replaced by cysts of variable size, with atresia of the ureter. **(b)** Renal ultrasound showing discrete cysts of variable size.

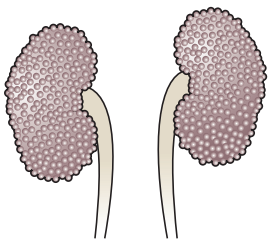


Figure 18.4 Autosomal recessive polycystic kidney disease (ARPKD). There is diffuse bilateral enlargement of both kidneys.

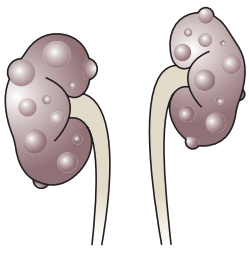


Figure 18.5 Autosomal dominant polycystic kidney disease (ADPKD). There are separate cysts of varying size between normal renal parenchyma. The kidneys are enlarged.

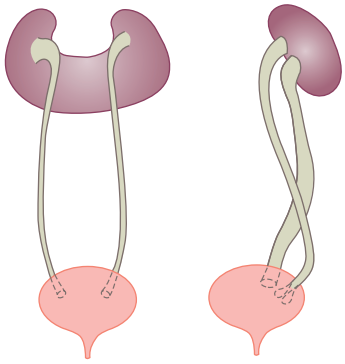


Figure 18.6 **(left)** Horseshoe kidney.
Figure 18.7 **(right)** Duplex kidney showing ureterocele of upper moiety and reflux into lower pole moiety.



Figure 18.8 Absent musculature syndrome (prune-belly syndrome). The name arises from the wrinkled appearance of the abdomen. It is associated with a large bladder, dilated ureters and cryptorchidism. (Courtesy of Dr Jane Deal.)

Urinary tract obstruction

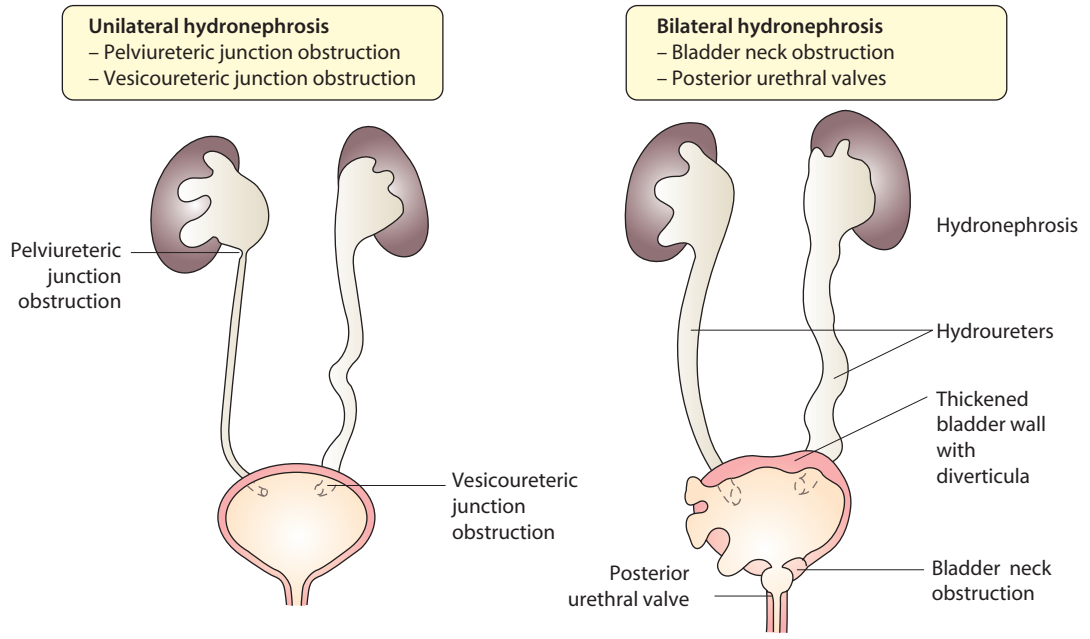


Figure 18.9a Obstruction to urine flow results in dilatation of the urinary tract proximal to the site of obstruction. Obstruction may be at the pelviureteric or vesicoureteric junction (left), the bladder neck or urethra (right).

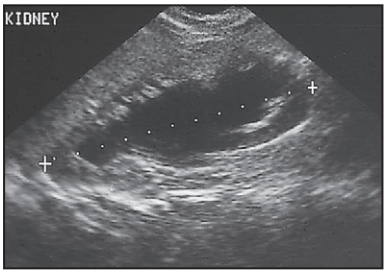


Figure 18.9b An ultrasound showing a dilated renal pelvis from pelviureteric junction obstruction.

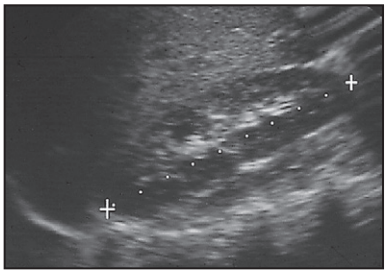


Figure 18.9c A normal ultrasound of the kidney is shown for comparison.

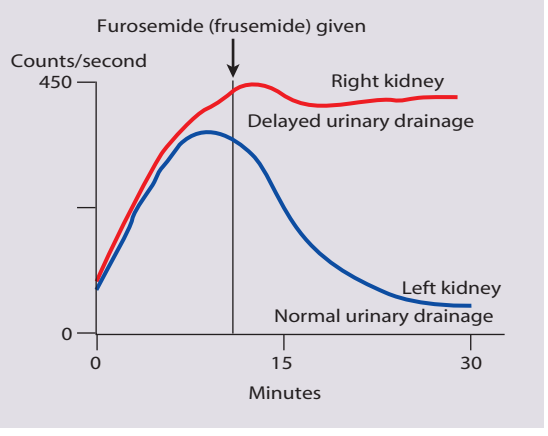


Figure 18.9d Graph from dynamic nuclear medicine scan MAG3 showing delayed excretion from a pelviureteric junction obstruction.

It is associated with several extra-renal features including cysts in liver and pancreas, cerebral aneurysms and mitral valve prolapse.

Abnormal caudal migration may result in a *pelvic kidney* or a *horseshoe kidney* (Fig. 18.6), when the lower poles are fused in the midline. The abnormal position may predispose to infection or obstruction to urinary drainage.

Premature division of the ureteric bud gives rise to a *duplex system*, which can vary from simply a bifid renal pelvis to complete division with two ureters. These ureters frequently have an abnormal drainage so that the ureter from the lower pole moiety often refluxes, whereas the upper pole ureter may drain ectopically into the urethra or vagina or may prolapse into the bladder (ureterocele) and urine flow may be obstructed (Fig. 18.7).

Failure of fusion of the infraumbilical midline structures results in exposed bladder mucosa (*bladder extrophy*). Absence or severe deficiency of the anterior abdominal wall muscles is frequently associated with a large bladder and dilated ureters (megacystis-megaureters) and cryptorchidism, the *absent musculature syndrome* (*prune-belly syndrome*) (Fig. 18.8).

Obstruction to urine flow may occur at the pelviureteric or vesicoureteric junction, at the *bladder neck* (e.g. due to disruption of the nerve supply, *neuropathic bladder*) or at the *posterior urethra* in a boy due to mucosal folds or a membrane, known as *posterior urethral valves*. The consequences of obstruction to urine flow are shown in Figures 18.9a-18.9d. At worst, this results in a *dysplastic kidney* which is small, poorly functioning and may contain cysts and aberrant embryonic tissue such as cartilage. In the most severe and bilateral cases Potter syndrome is present. Renal dysplasia can also occur in association with severe intrauterine vesicoureteric reflux, in isolation or in certain rare, inherited syndromes affecting multiple systems.

Antenatal treatment

The male fetus with posterior urethral valves may develop severe urinary outflow obstruction resulting in progressive bilateral hydronephrosis, poor renal growth and declining liquor volume with the potential to lead to pulmonary hypoplasia. Intrauterine bladder drainage procedures to prevent severe renal damage have been attempted but results have been disappointing. Early delivery is rarely indicated.

Postnatal management

An example of a protocol for infants with antenatally diagnosed anomalies is shown in Figure 18.10. Prophylactic antibiotics may be started at birth to try to prevent urinary tract infection, although practice varies between centres. As the newborn kidney has a low GFR, urine flow is low and mild outflow obstruction may not be evident in the first few days of life. The ultrasound scan should therefore be delayed for several weeks. However, bilateral hydronephrosis in a male infant warrants an ultrasound shortly after birth to exclude posterior urethral valves, which always requires urological intervention such as cystoscopic ablation (Case History 18.1).

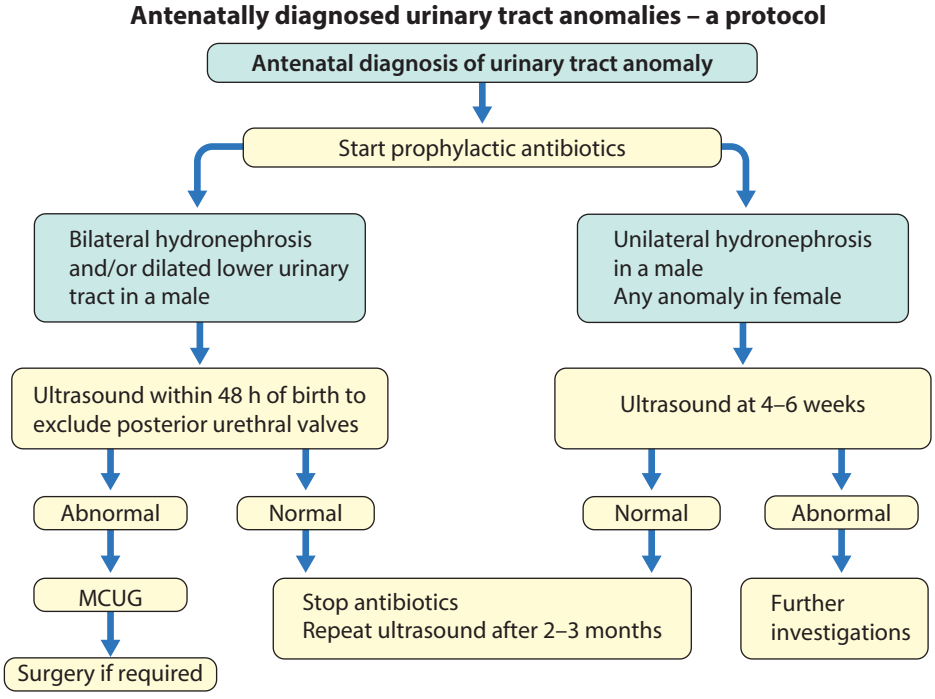


Figure 18.10 An example of a protocol for the management of infants with antenatally diagnosed urinary tract anomalies. (MCUG, micturating cystourethrogram.)

Case History

18.1 Posterior urethral valves

Bilateral hydronephrosis was noted on antenatal ultrasound at 20 weeks' gestation in a male fetus. There was poor renal growth, progressive hydronephrosis and decreasing volume of amniotic fluid (Fig. 18.11a) on repeated scans. After birth, prophylactic antibiotics were started. An urgent ultrasound showed bilateral hydronephrosis with small dysplastic kidneys. The bladder and ureters were grossly distended. The plasma creatinine was raised. A micturating cystourethrogram (MCUG) (Fig. 18.11b) showed vesicoureteric reflux, a dilated posterior urethra and posterior urethral valves which was treated endoscopically. Renal function initially improved but then progressed to chronic renal failure. He had a renal transplant at 10 years of age.

Bilateral hydronephrosis in a male infant requires urgent investigation to exclude posterior urethral valve.

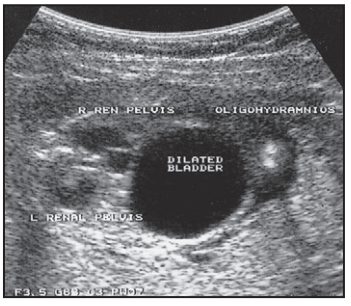


Figure 18.11a Antenatal ultrasound scan in an infant with urinary outflow obstruction from posterior urethral valve. (Courtesy of Mr Karl Murphy.)

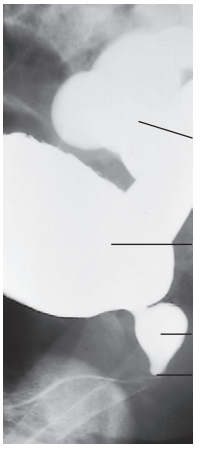


Figure 18.11b Micturating cystourethrogram (MCUG) in the same patient.

Urinary tract infection

About 3–7% of girls and 1–2% of boys have at least one symptomatic urinary tract infection (UTI) before the age of 6 years, and 12–30% of them have a recurrence within a year. UTI may involve the kidneys (pyelonephritis), when it is usually associated with fever and systemic involvement, or may be due to cystitis, when there may be no fever. UTI in childhood is important because:

- up to half of patients have a structural abnormality of their urinary tract
- pyelonephritis may damage the growing kidney by forming a scar, predisposing to hypertension and to chronic renal failure if the scarring is bilateral.

There are NICE guidelines on urinary tract infection in children, published in 2007, although they have proved to be controversial.

Box 18.1 Presentation of UTI in infants and children

Infants	Children
• Fever • Vomiting • Lethargy or irritability • Poor feeding/failure to thrive • Jaundice • Septicaemia • Offensive urine • Febrile convulsion (>6 months).	• Dysuria and frequency • Abdominal pain or loin tenderness • Fever with or without rigors (exaggerated shivering) • Lethargy and anorexia • Vomiting, diarrhoea • Haematuria • Offensive/cloudy urine • Febrile convulsion • Recurrence of enuresis.

Clinical features

Presentation of UTI varies with age (Box 18.1). In infants, symptoms are non-specific; fever is usually but not always present, and septicaemia may develop rapidly. The classical symptoms of dysuria, frequency and loin pain become more common with increasing age. Serious illness from septicaemia is described in the child with a fever in Chapter 14. Dysuria alone is usually due to cystitis, or vulvitis in girls or balanitis in uncircumcised boys. Symptoms suggestive of a UTI may also occur following sexual abuse.

Collection of samples

The commonest error in the management of UTI in children, and especially in infants, is failure to establish

Table 18.3 Methods and interpretation of dipstick testing in children

Methods of dipstick testing	
Nitrite stick testing	Positive result useful as very likely to indicate a true UTI But some children with a UTI are nitrite-negative
Leucocyte esterase stick testing (for WBCs)	May be present in children with UTI but may also be negative Present in children with febrile illness without UTIs Positive in balanitis and vulvovaginitis
Interpretation of results	
Leucocyte esterase and nitrite positive	Regard as UTI
Leucocyte esterase negative and nitrite positive	Start antibiotic treatment Diagnosis depends on urine culture
Leucocyte esterase positive and nitrite negative	Only start antibiotic treatment if clinical evidence of UTI Diagnosis depends on urine culture
Leucocyte esterase and nitrite negative	UTI unlikely. Repeat or send urine for culture if clinical history suggests UTI
Blood, protein, and glucose present on stick testing	Useful in any unwell child to identify other diseases, e.g. nephritis, diabetes mellitus, but will not discriminate between children with and without UTIs

the diagnosis properly in the first place. If the diagnosis of a UTI is not made, the opportunity to prevent renal damage may be missed, or, if incorrectly diagnosed, may lead to unnecessary invasive investigations.

For the child in nappies, urine can be collected by:

- A 'clean-catch' sample into a waiting clean pot when the nappy is removed. This is the recommended method
- An adhesive plastic bag applied to the perineum after careful washing, although there may be contamination from the skin
- A urethral catheter if there is urgency in obtaining a sample and no urine has been passed
- Suprapubic aspiration (SPA), when a fine needle attached to a syringe is inserted directly into the bladder just above the symphysis pubis under ultrasound guidance; it may be used in severely ill infants requiring urgent diagnosis and treatment, but it is an invasive procedure, and is increasingly replaced by urethral catheter sampling.

In the older child, urine can be obtained by collecting a midstream sample. Careful cleaning and collection are necessary, as contamination with both white cells and bacteria can occur from under the foreskin in boys, and from reflux of urine into the vagina during voiding in girls.

Ideally, the urine sample should be microscoped to identify organisms and cultured straight away. This is indicated in all infants and children <3 years old with a suspected UTI. If this is not possible, the urine sample should be refrigerated to prevent the overgrowth of contaminating bacteria. Urinary white cells are not a reliable feature of a UTI, as they may lyse during storage and may be present in febrile children without a UTI and in children with balanitis or vulvovaginitis. Dipsticks can be used as a screening test. Urine culture should still be performed unless both leucocyte esterase and nitrite are negative or if the clinical symptoms and dipstick tests do not correlate (Table 18.3).

A bacterial culture of $>10^5$ colony-forming units of a single organism per millilitre in a properly collected specimen gives a 90% probability of infection. If the same result is found in a second sample, the probability rises to 95%. A growth of mixed organisms usually represents contamination, but if there is doubt, another sample should be collected. Any bacterial growth of a single organism per millilitre in a catheter sample or suprapubic aspirate is considered diagnostic of infection.

A urine sample should be tested in all infants with an unexplained fever >38°C.

Bacterial and host factors that predispose to infection

Infecting organism

UTI is usually the result of bowel flora entering the urinary tract via the urethra, except in the newborn when it is more likely to be haematogenous. The commonest organism is *E. coli*, followed by *Klebsiella*, *Proteus* and *Pseudomonas* and *Strep. faecalis*. *Proteus* infection is more commonly diagnosed in boys than in girls, possibly because of its presence under the prepuce. *Proteus* infection predisposes to the formation of phosphate stones by splitting urea to ammonia and thus alkalinising the urine. *Pseudomonas* infection may indicate the presence of some structural abnormality in the urinary tract affecting drainage.

Antenatally diagnosed renal or urinary tract abnormality

Increases risk of infection.

Incomplete bladder emptying

Contributing factors in some children are:

- Infrequent voiding, resulting in bladder enlargement
- Vulvitis
- Incomplete micturition with residual post-micturition bladder volumes
- Obstruction by a loaded rectum from constipation
- Neuropathic bladder
- Vesicoureteric reflux.

Vesicoureteric reflux

Vesicoureteric reflux (VUR) is a developmental anomaly of the vesicoureteric junctions. The ureters are displaced laterally and enter directly into the bladder rather than at an angle, with a shortened or absent intramural course. Severe cases may be associated with renal dysplasia. It is familial, with a 30–50% chance of occurring in first-degree relatives. It may also occur with bladder pathology, e.g. a neuropathic bladder or urethral obstruction, or temporarily after a UTI. Its severity varies from reflux into the lower end of an undilated ureter during micturition to the severest form with reflux during bladder filling and voiding, with a distended ureter, renal pelvis and clubbed calyces (Fig. 18.12). Mild reflux is unlikely to be of significance, but the more severe degrees of VUR may be associated with *intrarenal reflux* (IRR), the backflow of urine from the renal pelvis into the papillary collecting ducts; intrarenal reflux is associated with a particularly high risk of renal scarring if UTIs occur. The incidence of renal defects increases with increasing severity of reflux. There is considerable controversy as to whether renal scarring is a congenital abnormality already present in children with reflux and which predisposes to infection or if children with reflux have normal kidneys at birth which are damaged by UTIs and that preventing UTIs in these children prevents scars. Reflux tends to resolve with age especially lower grades of VUR.

Reflux with associated ureteric dilatation is important, as:

- urine returning to the bladder from the ureters after voiding results in incomplete bladder emptying, which encourages infection
- the kidneys may become infected (pyelonephritis), particularly if there is intrarenal reflux
- bladder voiding pressure is transmitted to the renal papillae; this may contribute to renal damage if voiding pressures are high.

Infection may destroy renal tissue, leaving a scar, resulting in a shrunken, poorly functioning segment of kidney (reflux nephropathy). If scarring is bilateral and severe, chronic renal failure may develop. The risk for hypertension in childhood or early adult life is variously estimated to be up to 10%.

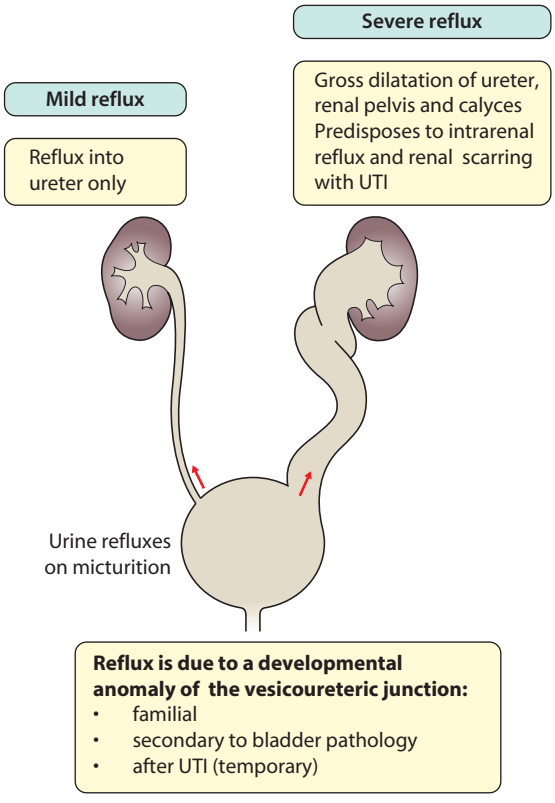


Figure 18.12 Vesicoureteric reflux.

Investigation

The extent to which a child with a UTI should be investigated is controversial. This is not only because of the invasive nature and radiation burden of the tests but also because of the lack of an evidence base to show that outcome is improved (unless urinary obstruction is demonstrated). Mild reflux usually resolves spontaneously, and operative intervention to stop reflux has not been shown to decrease renal damage. Furthermore, there is no evidence that antibiotic prophylaxis is any better than prompt treatment. There has, therefore, been a move away from extensive investigation of all children with UTIs to those who have had atypical or recurrent UTIs. Atypical UTI includes:

- Seriously ill or septicaemia
- Poor urine flow
- Abdominal or bladder mass
- Raised creatinine
- Failure to respond to suitable antibiotics within 48 h
- infection with non-*E. coli* organism.

An initial ultrasound will identify:

- Serious structural abnormalities and urinary obstruction
- Renal defects (but poor at identifying renal scars).

Subsequent investigations will depend on the results of the ultrasound. The need for any investigations in a child with only bladder symptoms (lower urinary tract infection/cystitis) is also controversial. If urethral obstruction is suspected (abnormal bladder in a boy),

Case History

18.2 Urinary tract infection

Jack, a 2-month-old infant, stopped feeding and had a high, intermittent fever. He was referred to hospital, where he had an infection screen. Urine examination showed >100 white blood cells, $>10^5$ *E. coli*/ml. He was treated with intravenous antibiotics. An ultrasound showed that the left kidney was smaller than the right kidney with dilated ureters. He was started on prophylactic antibiotics. A DMSA scan (Fig. 18.14) performed 3 months later confirmed bilateral renal scarring, with the left kidney contributing 33% of renal function. The MCUG (Fig. 18.15) showed bilateral vesicoureteric reflux. At 4 years of age, the reflux had resolved and antibiotic prophylaxis was stopped. His blood pressure and renal growth and function continue to be monitored.

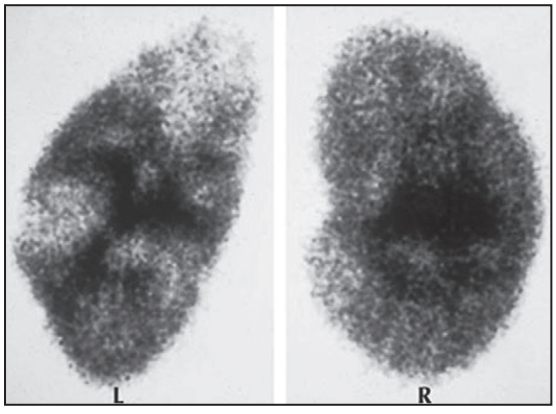


Figure 18.14 DMSA scan showing bilateral renal scarring, more severe on left upper pole.

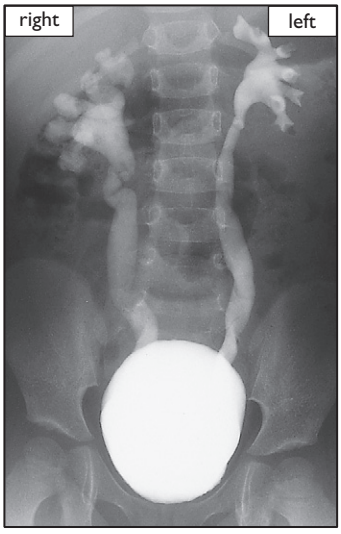


Figure 18.15 Micturating cystourethrogram showing bilateral vesicoureteric reflux with ureteric dilatation and dilated, clubbed calyces on the right.

First urinary tract infection - a protocol for initial management and investigation

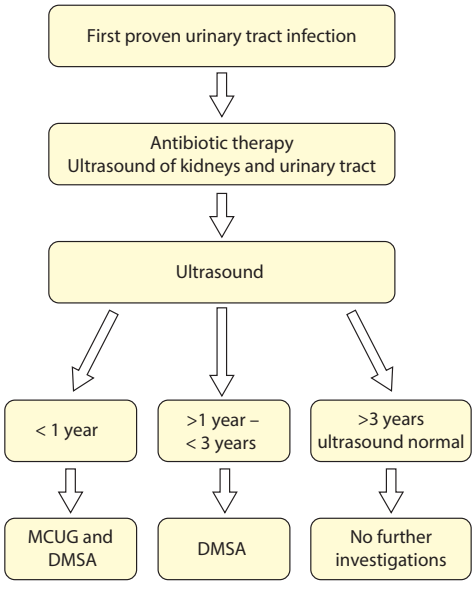


Figure 18.13 An example of a protocol for the initial management and investigation of a first urinary tract infection. This is a controversial area. The UK NICE guidelines, 2007 do not recommend ultrasound examination for first UTI if there was response to antibiotic treatment within 48 h, unless <6 months old or atypical or recurrent, but many paediatric nephrologists consider this approach too conservative and follow protocols like the one shown here.

MCUG should be performed promptly. Functional scans should be deferred for 3 months after a UTI, unless the ultrasound is suggestive of obstruction, to avoid missing a newly developed scar and because of false-positive results from transient inflammation. Medical measures for the prevention of UTI should be initiated.

A suggested schema for investigation of the first proven UTI is shown in Figure 18.13, but varies between centres.

Management

All infants <3 months old with suspicion of a UTI or if seriously ill should be referred immediately to a hospital. They require intravenous antibiotic therapy (e.g. cefotaxime) until the temperature has settled, when oral treatment is substituted (see Case History 18.2.)

Infants >3 months and children with acute pyelonephritis/upper urinary tract infection (bacteriuria and fever ≥38°C or bacteriuria and loin pain/tenderness even if fever is <38°C) are usually treated with oral antibiotics with low resistance patterns (e.g. co-amoxiclav for 7–10 days); or else intravenous antibiotics, e.g. cefotaxime is given for 2–4 days followed by oral antibiotics for a total of 7–10 days. The choice of antibiotic is adjusted according to sensitivity on urine culture.

Children with cystitis/lower urinary tract infection (dysuria but no systemic symptoms or signs) can be treated with oral antibiotics for 3 days.

Medical measures for the prevention of UTI

The aim is to ensure washout of organisms that ascend into the bladder from the perineum; and to reduce the presence of aggressive organisms in the stool, perineum and under the foreskin:

- High fluid intake to produce a high urine output
- Regular voiding
- Ensuring complete bladder emptying by encouraging the child to try a second time to

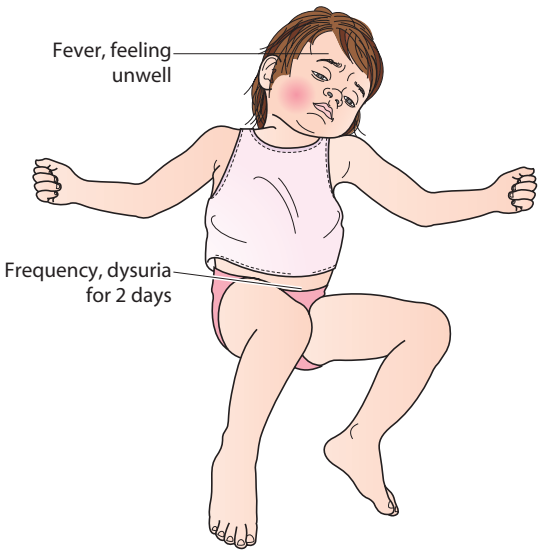
Summary

A child with a first urinary tract infection

Why important?
Up to half have a structural abnormality of their urinary tract
Pyelonephritis may damage the growing kidney by forming a renal scar, which may result in hypertension and chronic renal failure

Diagnosis secure?
- Suggestive clinical features?
- Upper or lower urinary tract infection?
- Urine sample properly collected and processed?
- Culture of single organism >10^5/ml if clean catch or mid-stream urine or else any organisms on suprapubic aspirate or catheter sample?

Predisposing factors?
Incomplete bladder emptying
Constipation
Vesicoureteric reflux



Why investigate?
To identify serious structural abnormalities, urinary obstruction, renal scars, vesicoureteric reflux.

What investigation?
Consider:
- Ultrasound of kidneys and urinary tract
- DMSA to check for renal scars 3 months after UTI
- MAG3 or MCUG to detect obstruction and vesicouretic reflux.

Management
Treat infection with antibiotics
Advice about medical preventative measures:
- High fluid intake
- Regular voiding, double micturition
- Prevent or treat constipation
- Good perineal hygiene
- *Lactobacillus acidophilus*

Advise to check urine culture if develops clinical features suggestive of non-specific illness

If renal scarring or reflux on investigation, or develops recurrent UTIs:
- Consider low-dose antibiotic prophylaxis
- Monitor blood pressure, renal growth and function

empty his bladder after a minute or two, commonly known as double micturition; this empties any urine residue or refluxed urine returning to the bladder
- Prevention or treatment of constipation
- Good perineal hygiene
- *Lactobacillus acidophilus*, a probiotic to encourage colonisation of the gut by this organism and reduce the number of pathogenic organisms that might potentially cause invasive disease
- Antibiotic prophylaxis, although this is controversial. It is often used in those under 2 years of age with a congenital abnormality of the kidneys or urinary tract or who have had an upper urinary tract infection and those with severe reflux. Trimethoprim (2 mg/kg at night) is used most often, but nitrofurantoin or cephalexin may be given. Broad-spectrum, poorly absorbed antibiotics such as amoxicillin should be avoided.

Follow-up of children with recurrent UTIs, renal scarring or reflux

In these children:

- Urine culture should be checked with any non-specific illness in case it is caused by a UTI (urine should not be cultured routinely).
- Long-term, low-dose antibiotic prophylaxis can be used. There is no evidence for when antibiotic prophylaxis should be stopped.
- Circumcision in boys may sometimes be considered, as there is evidence that it reduces the incidence of urinary tract infection.
- Anti-reflux surgery may be indicated if there is progression of scarring with ongoing reflux, but it has not been shown to improve outcome.
- Blood pressure should be checked annually if renal defects are present.
- Regular assessment of renal growth and function is necessary if there are bilateral defects because of the risk of chronic renal failure.

If there are further symptomatic UTIs in younger children, investigations are required to determine whether there are new scars or continuing reflux.

Enuresis

Primary nocturnal enuresis

This is considered in Chapter 23.

Daytime enuresis

This is a lack of bladder control during the day in a child old enough to be continent (over the age of 3–5 years). Nocturnal enuresis is also usually present. It may be caused by:

- Lack of attention to bladder sensation: a manifestation of a developmental or psychogenic problem, although it may occur in otherwise normal children who are too preoccupied with what they are doing to respond to the sensation of a full bladder
- Detrusor instability (sudden, urgent urge to void induced by sudden bladder contractions)
- Bladder neck weakness
- A neuropathic bladder (bladder is enlarged and fails to empty properly, irregular thick wall and is associated with spina bifida and other neurological conditions)
- A urinary tract infection (rarely in the absence of other symptoms)
- Constipation
- An ectopic ureter, causes constant dribbling and child is always damp.

Examination may reveal evidence of a neuropathic bladder, i.e. the bladder may be distended, there may be abnormal perineal sensation and anal tone or abnormal leg reflexes and gait. Sensory loss in the distribution of the S2, 3 and 4 dermatomes should be sought. A spinal lesion may be present. Girls who are dry at night but wet on getting up are likely to have pooling of urine from an ectopic ureter opening into the vagina.

A urine sample should be examined for microscopy, culture and sensitivity. Other investigations are performed if indicated. An ultrasound may show bladder pathology, with incomplete bladder emptying or thickening of the bladder wall. Urodynamic studies may be required. An X-ray of the spine may reveal a vertebral anomaly. An MRI scan may be required to confirm or exclude a non-bony spinal defect such as tethering of the cord.

Affected children in whom a neurological cause has been excluded may benefit from star charts, bladder training and pelvic floor exercises. Constipation should be treated. A small portable alarm with a pad in the pants, which is activated by urine, can be used when there is lack of attention to bladder sensation. Anticholinergic drugs, such as oxybutynin, to damp down bladder contractions, may be helpful if other measures fail.

Secondary (onset) enuresis

The loss of previously achieved urinary continence may be due to:

- Emotional upset, the commonest cause
- UTI
- Polyuria from an osmotic diuresis in diabetes mellitus or a renal concentrating disorder, e.g. sickle cell disease or chronic renal failure.

Investigation should include:

- Testing a urine sample for infection, glycosuria and proteinuria
- Assessment of urinary concentrating ability by measuring the osmolality of an early morning urine sample. Rarely, a formal water deprivation test may be needed to exclude a urinary concentrating defect
- Ultrasound of the renal tract.